



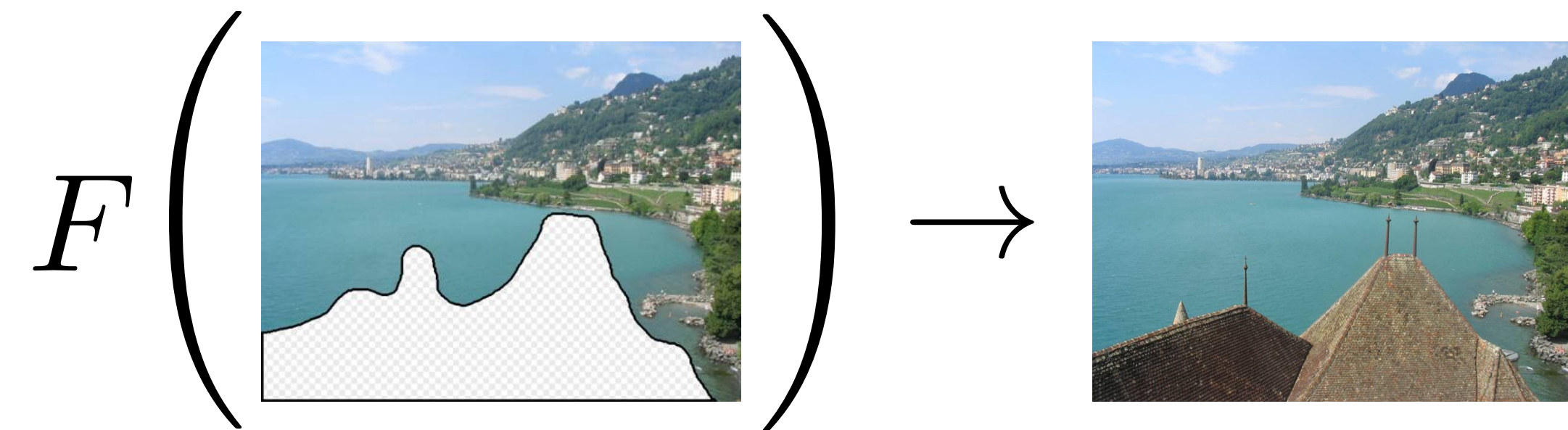
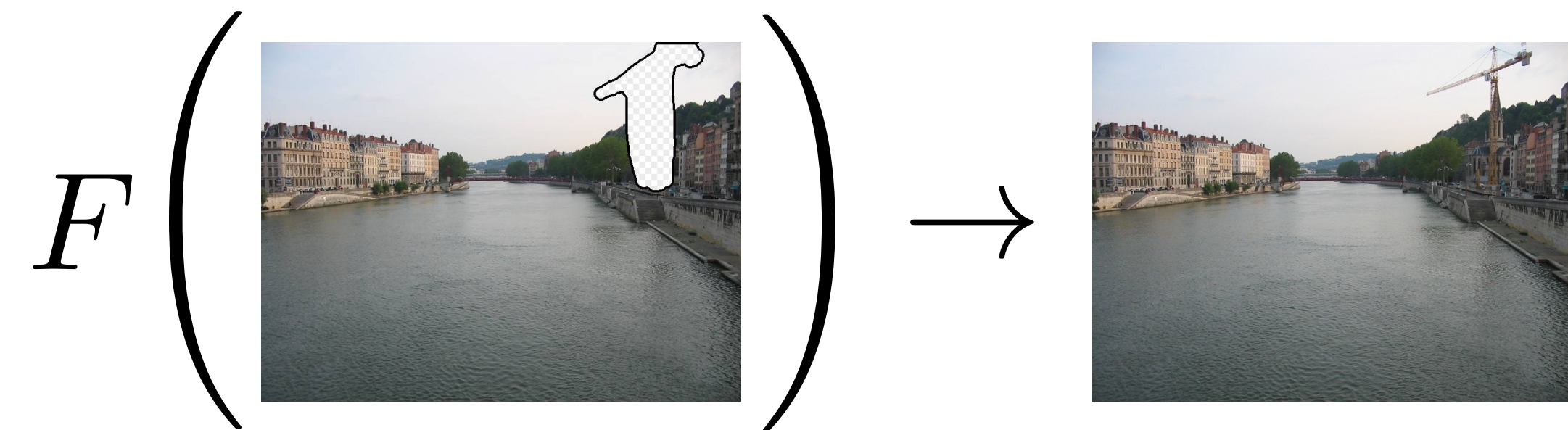
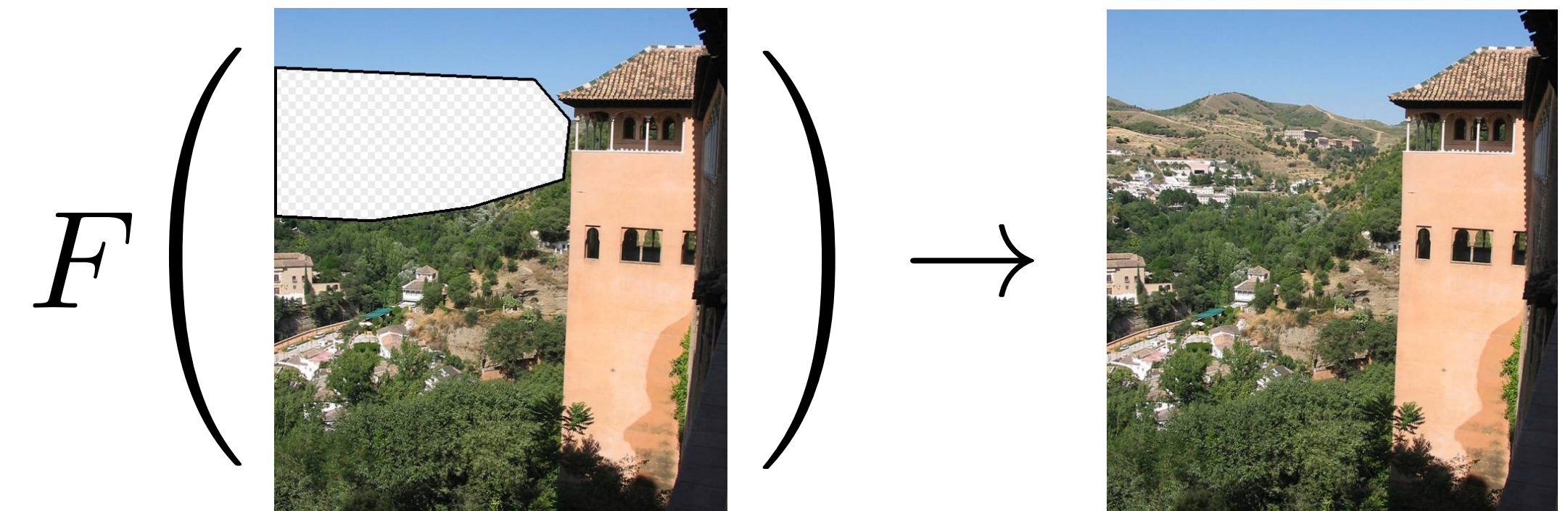
BITS Pilani
Pilani | Dubai | Goa | Hyderabad

Computer Vision

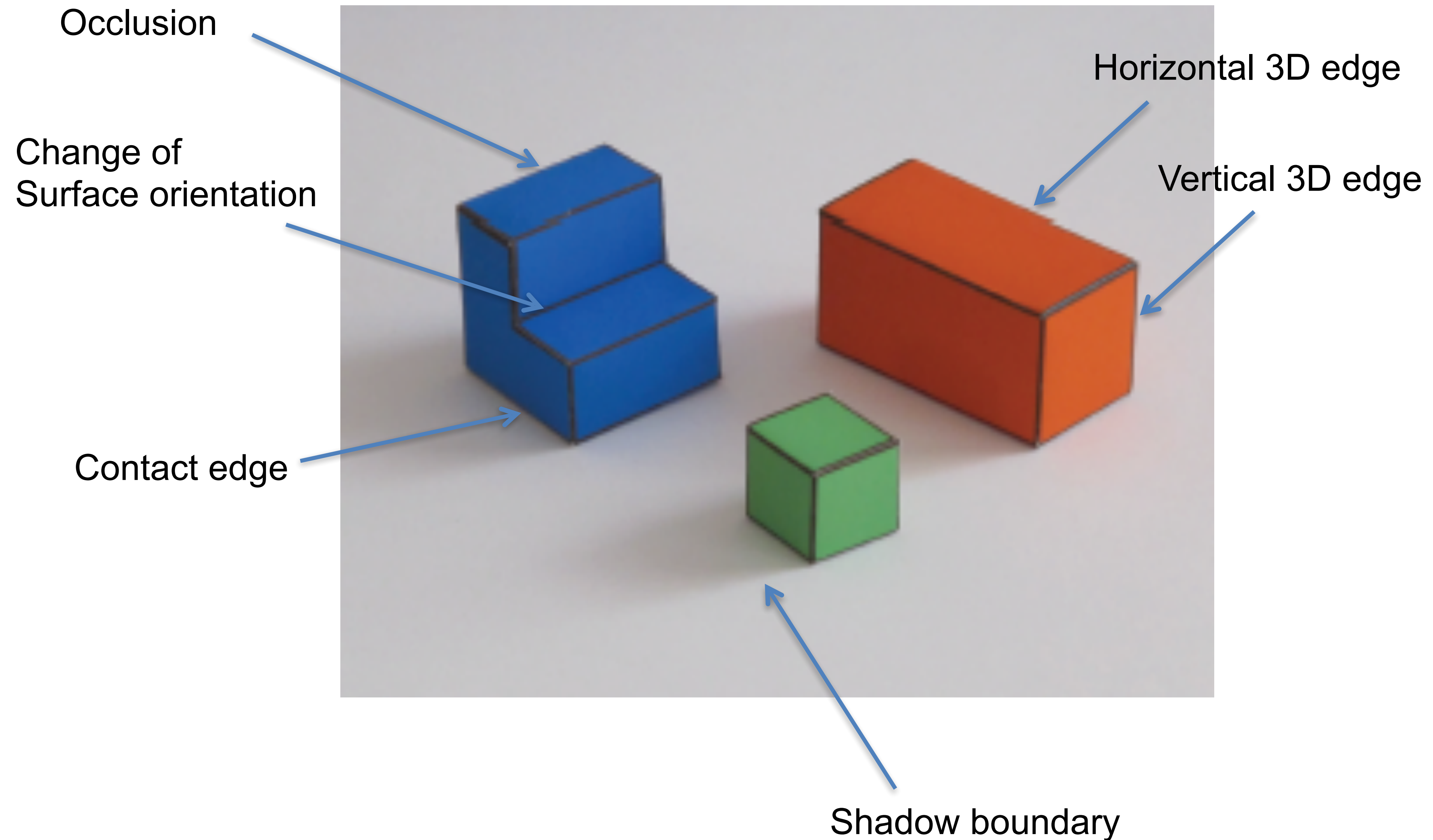
S2_25_AIML*ZG525, M.Tech (AIML)

Session #8/9:
Recognition

Raghavendra Singh

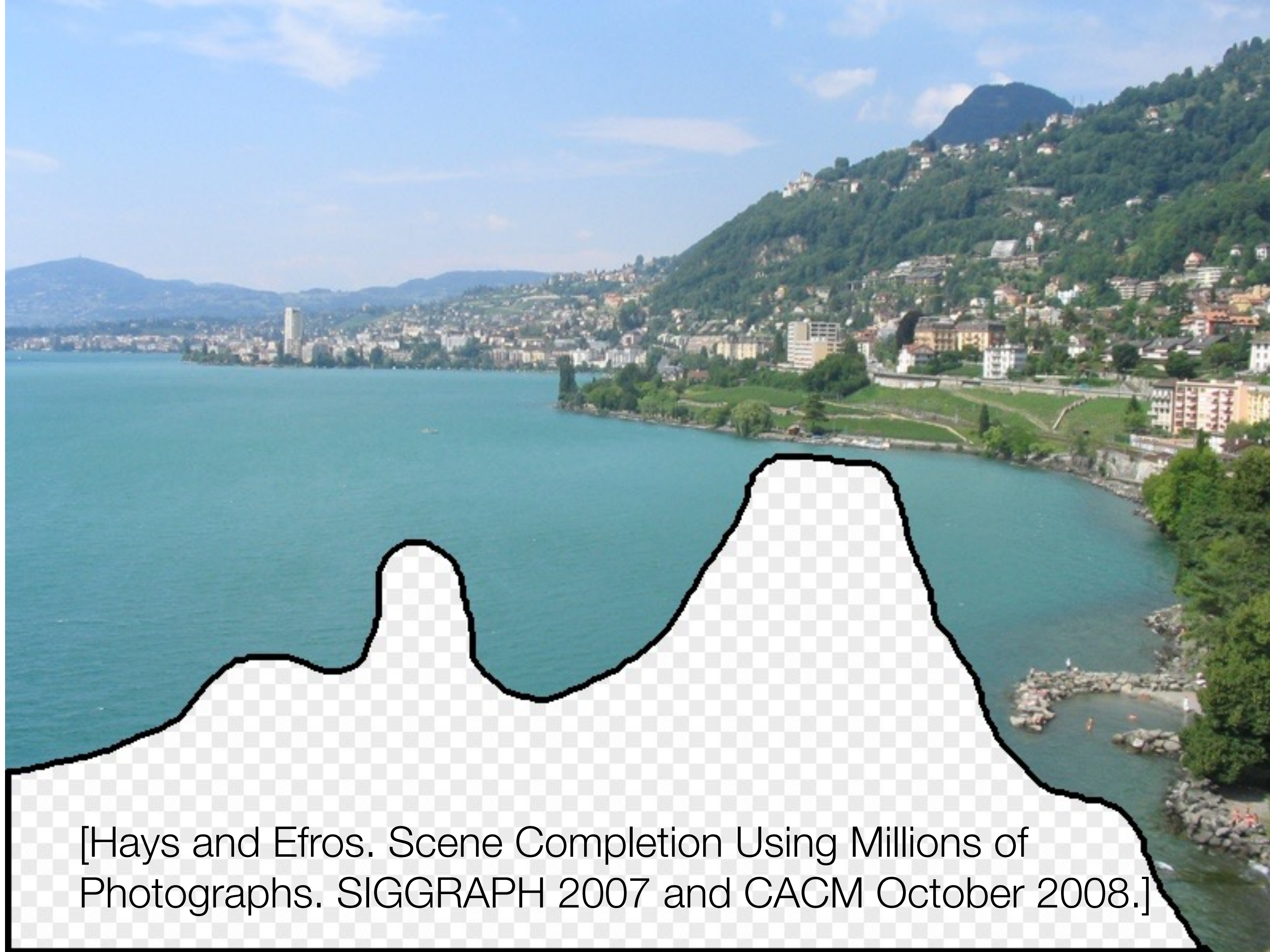


Old method of solving vision: hand-coded by humans.
Understandable, but not robust.





[Slide credit: Alexei Efros]



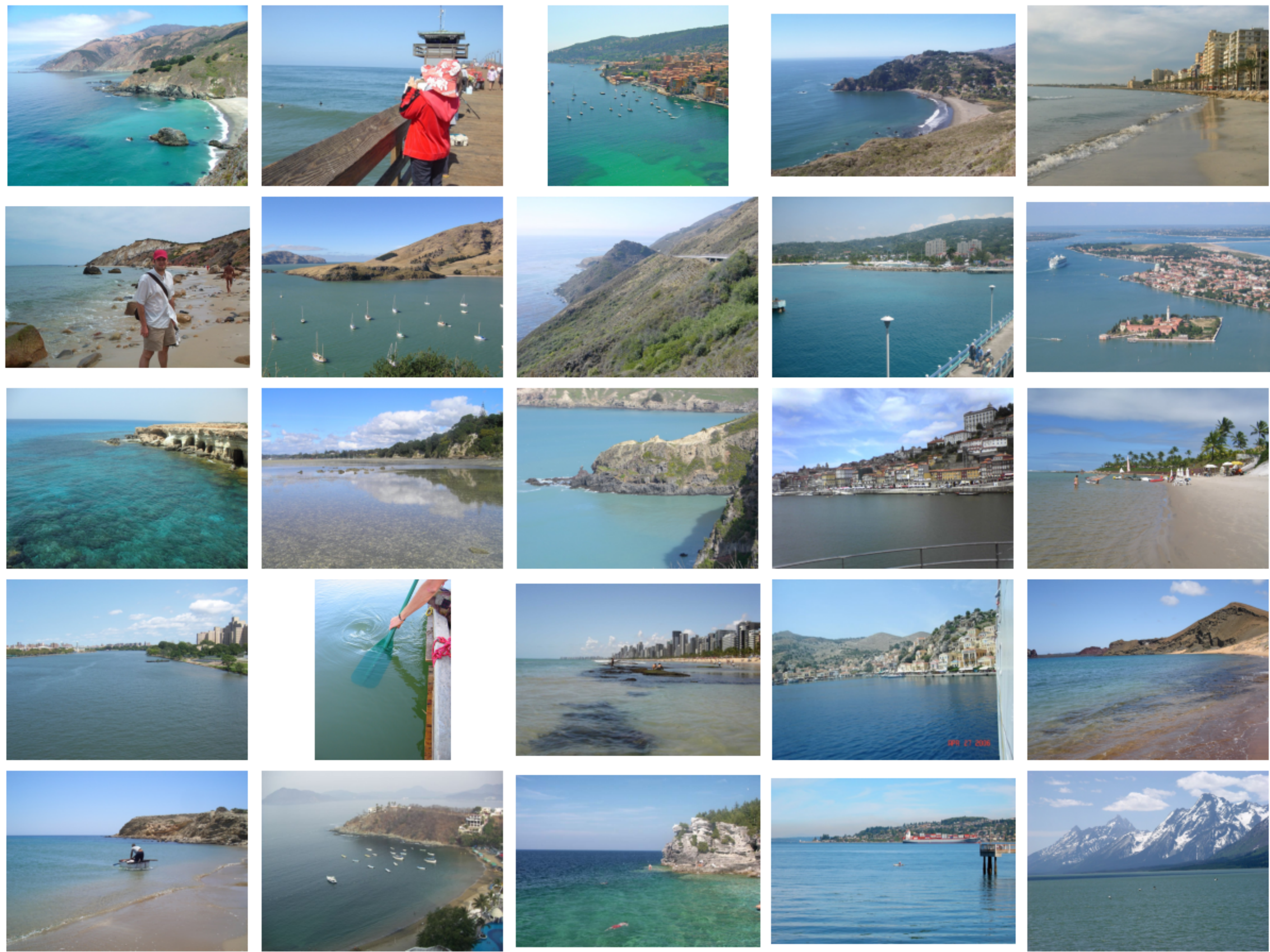
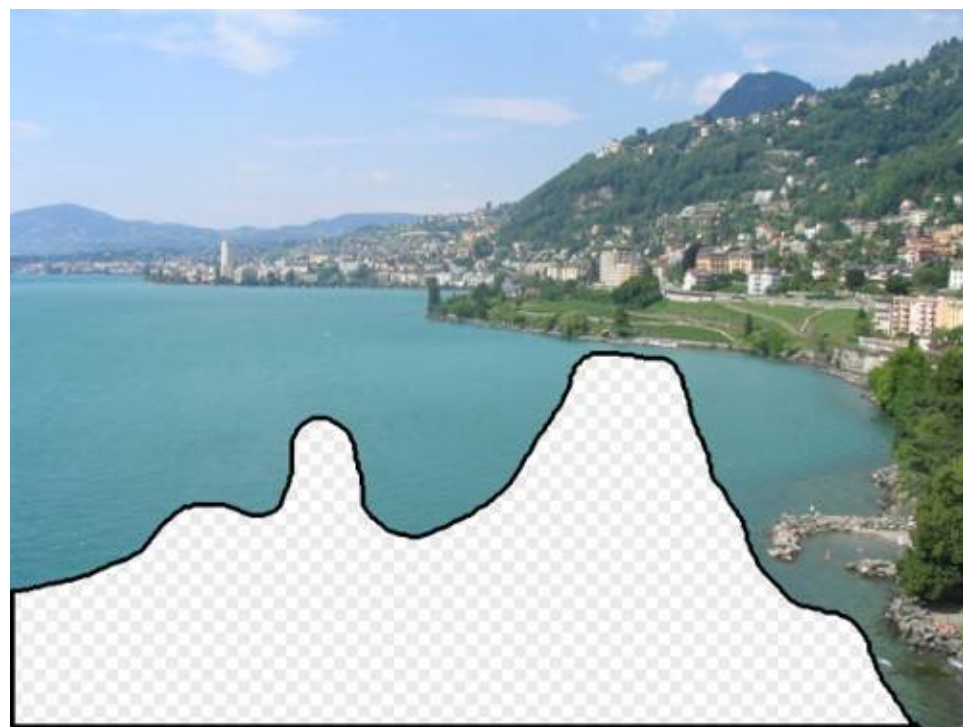
[Hays and Efros. Scene Completion Using Millions of Photographs. SIGGRAPH 2007 and CACM October 2008.]

[Slide credit: Alexei Efros]

2 Million Flickr Images

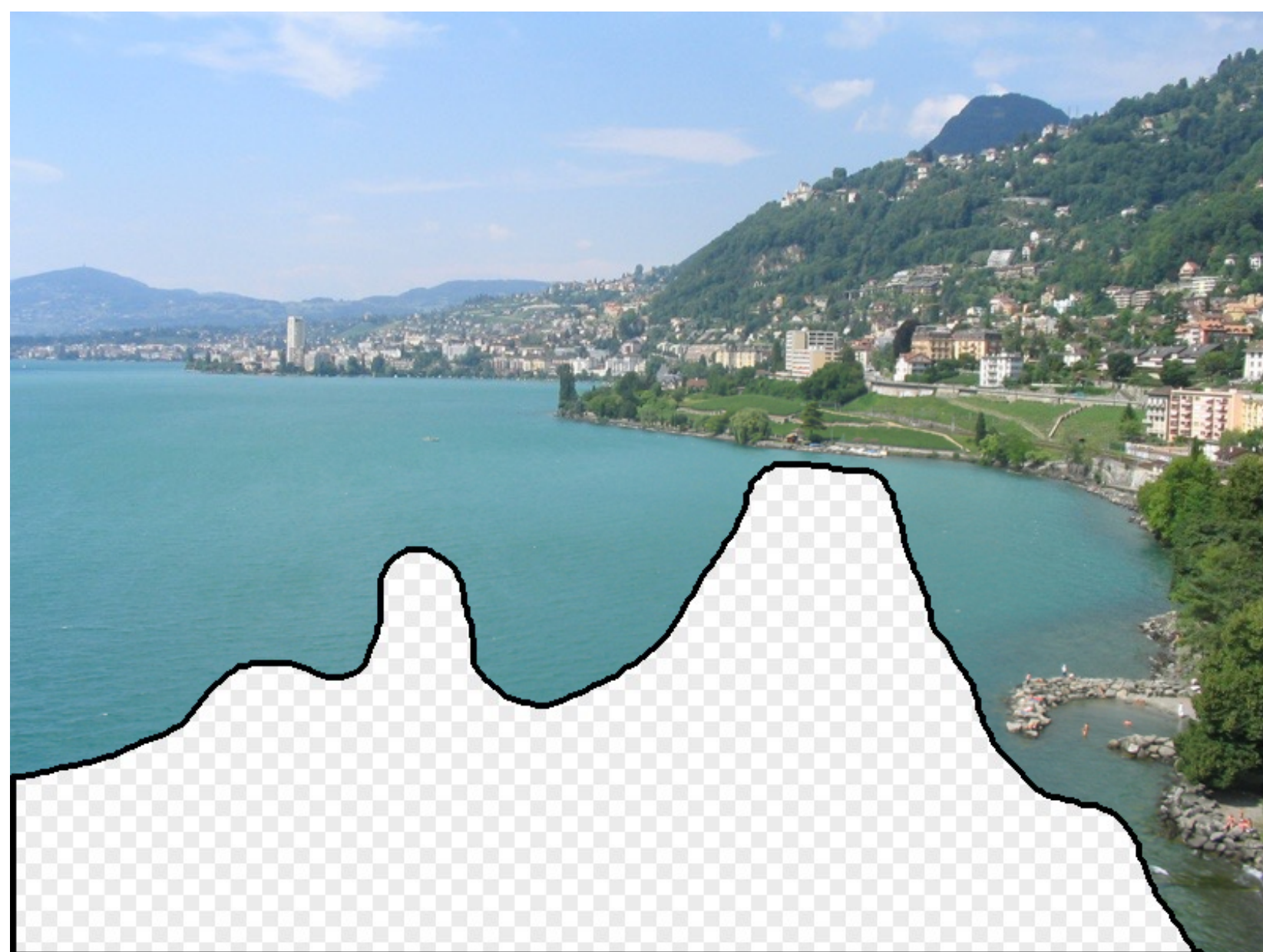


[Slide credit: Alexei Efros]



... 200 total

[Slide credit: Alexei Efros]



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[Slide credit: Alexei Efros]



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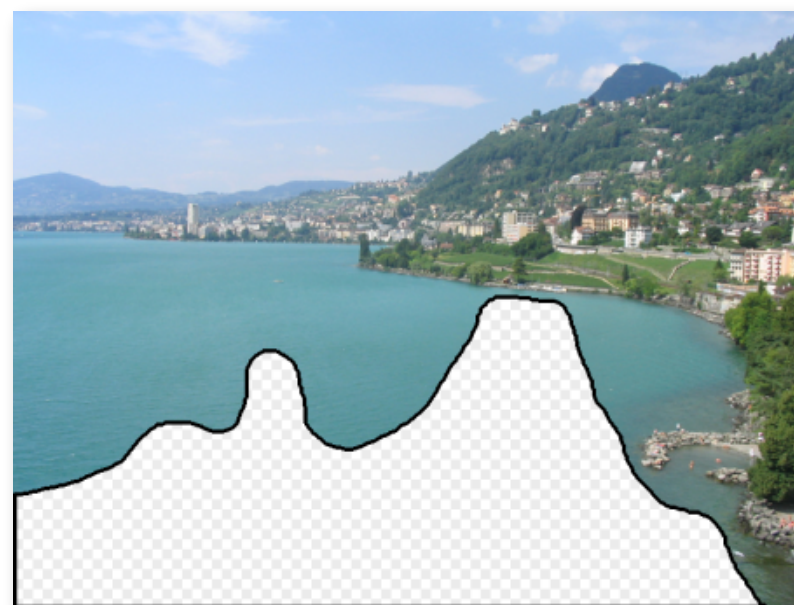


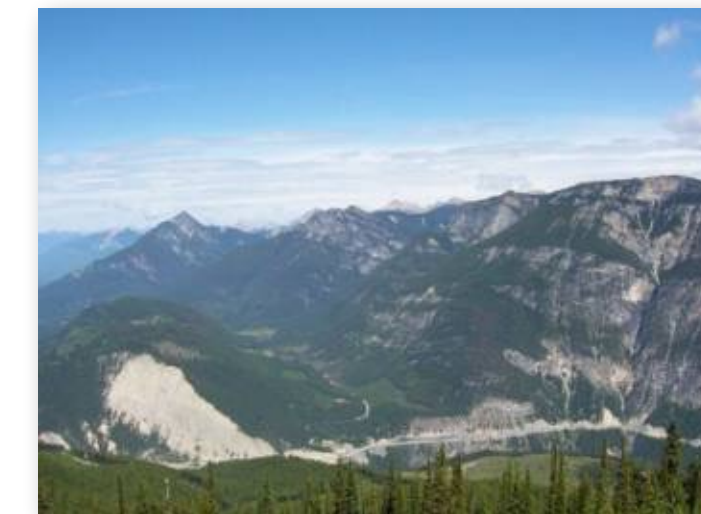
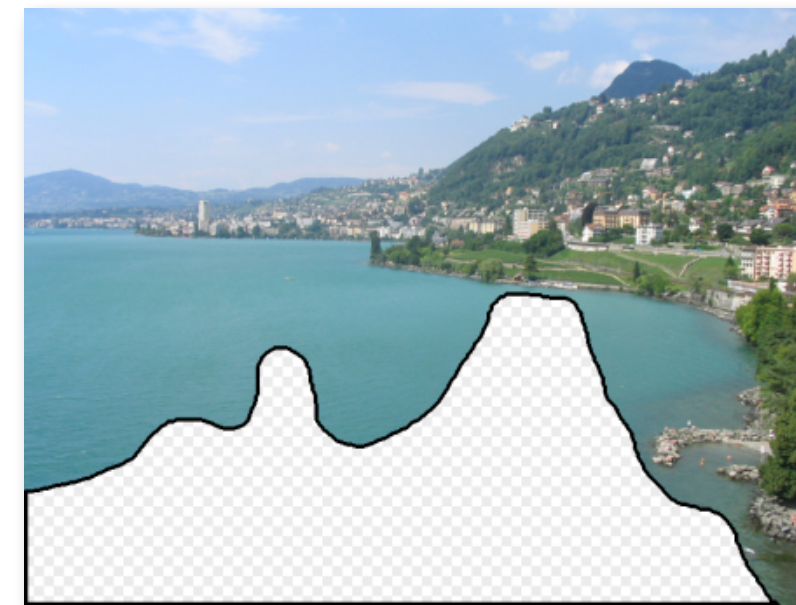
[Slide credit: Alexei Efros]



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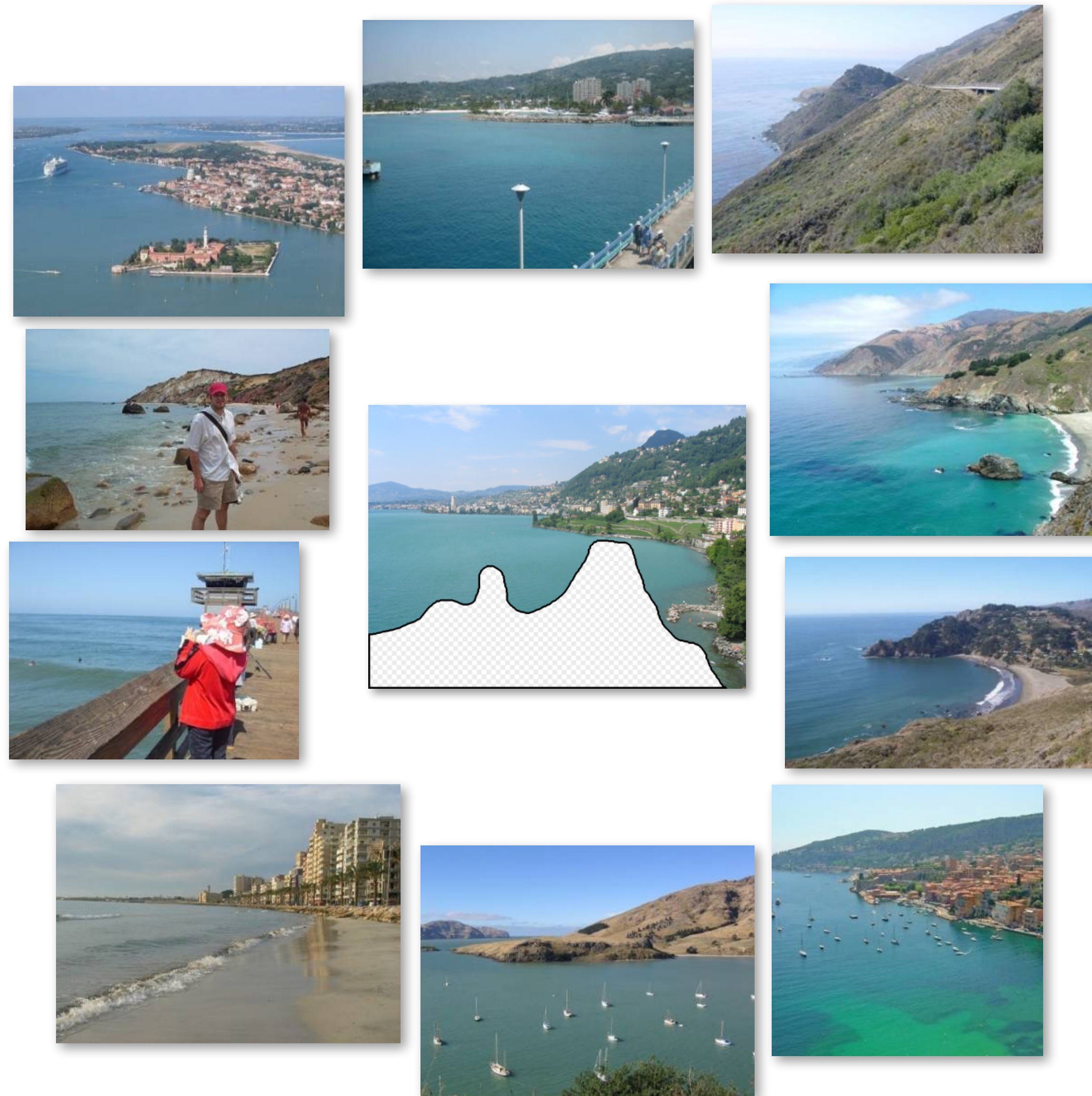
Why does it work?





Nearest neighbors from a
collection of 20 thousand images

[Slide credit: Alexei Efros]



Nearest neighbors from a
collection of 2 million images

[Slide credit: Alexei Efros]

“Unreasonable Effectiveness of Data”

[Halevy, Norvig, Pereira 2009]

Parts of our world can be explained by elegant mathematics

physics, chemistry, astronomy, etc.

But much cannot

psychology, economics, genetics, etc.

Enter The Data!

Great advances in several fields:

e.g., speech recognition, machine translation

Case study: Google

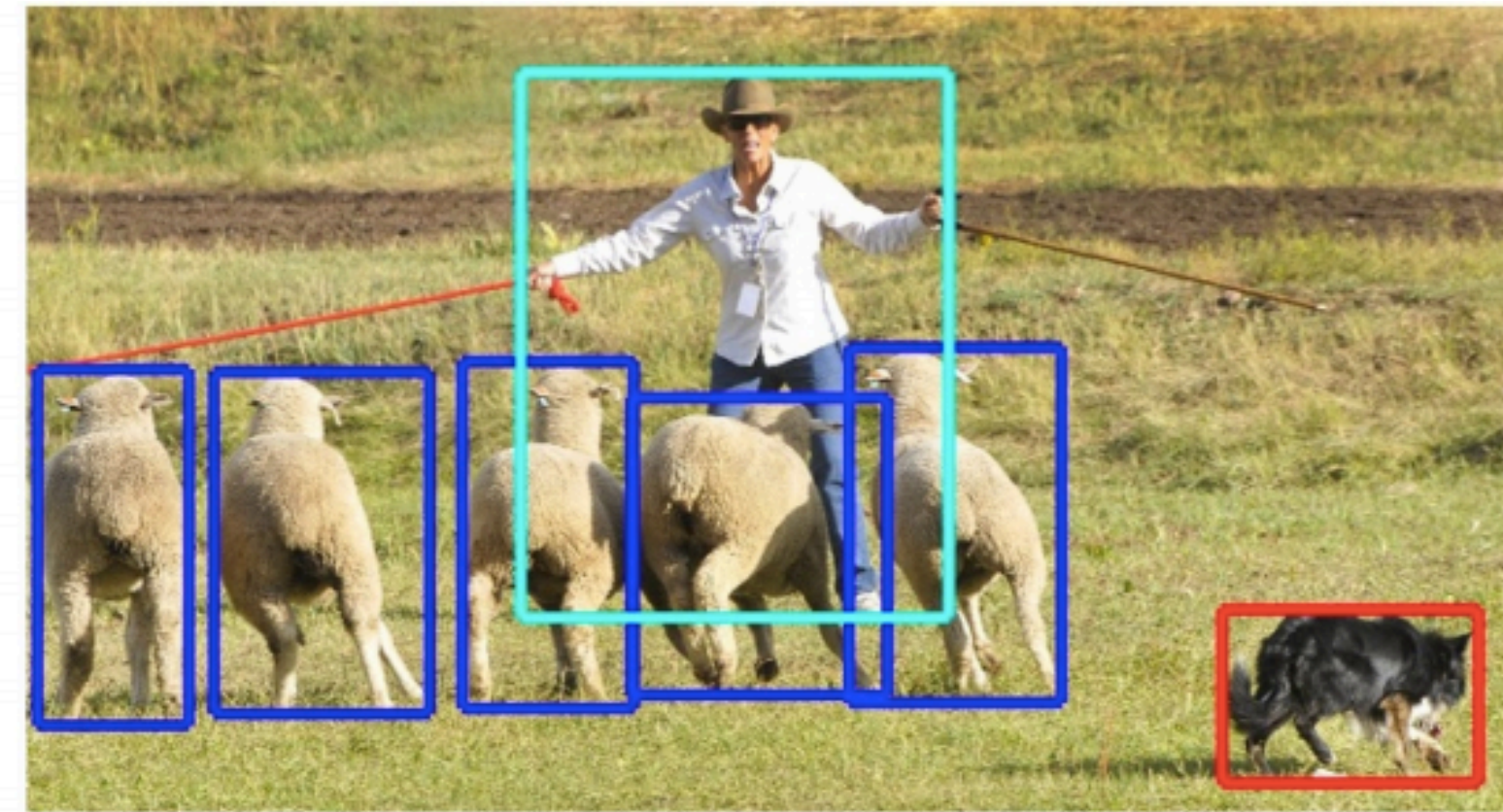
“For many tasks, once we have a billion or so examples, we essentially have a closed set that represents (or at least approximates) what we need...”

[Slide credit: Alexei Efros]

Detection, semantic segmentation, instance segmentation



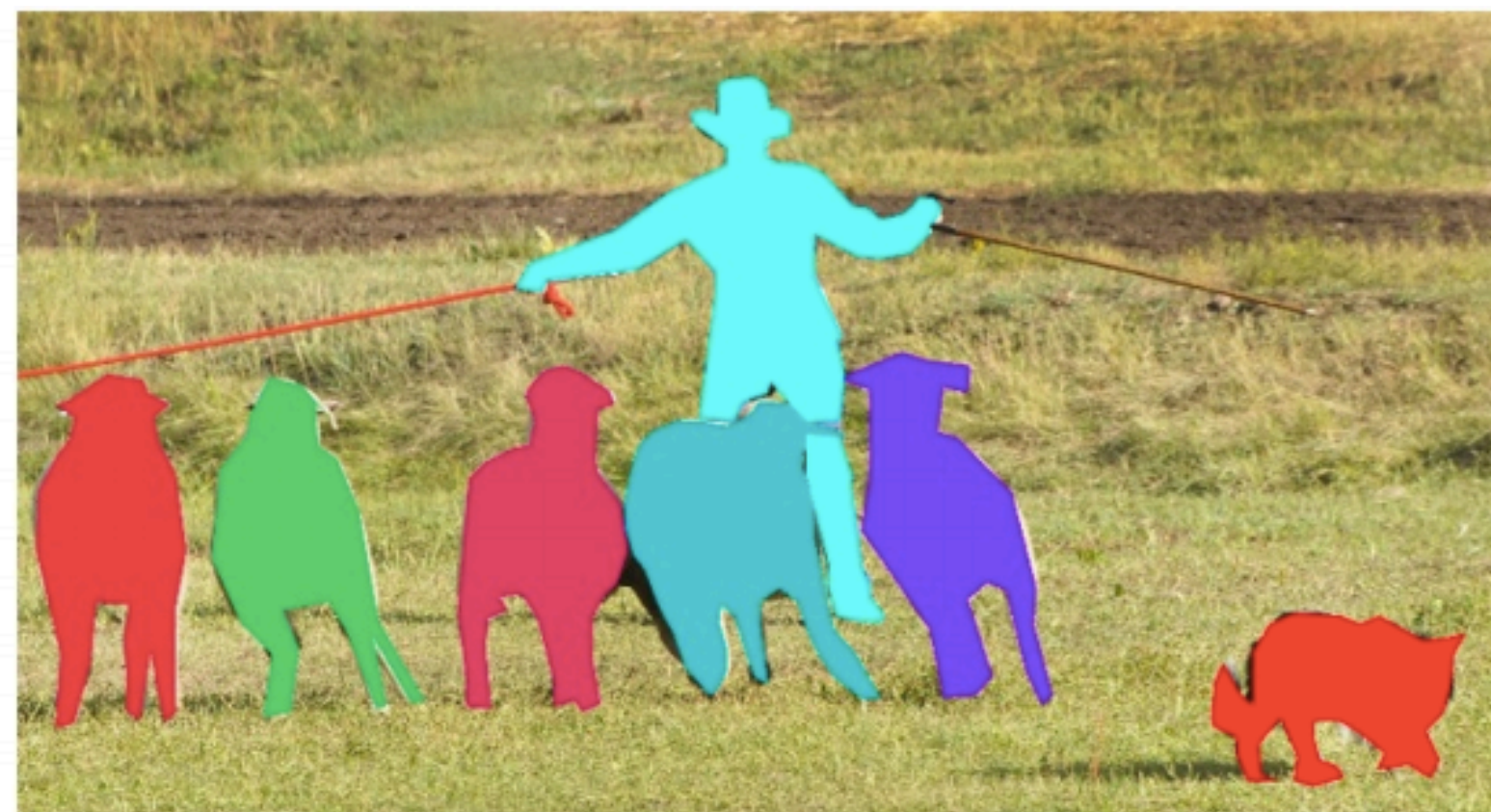
image classification



object detection



semantic segmentation

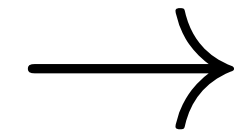


instance segmentation

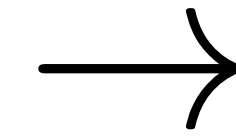
Framework for learning from data

Learning

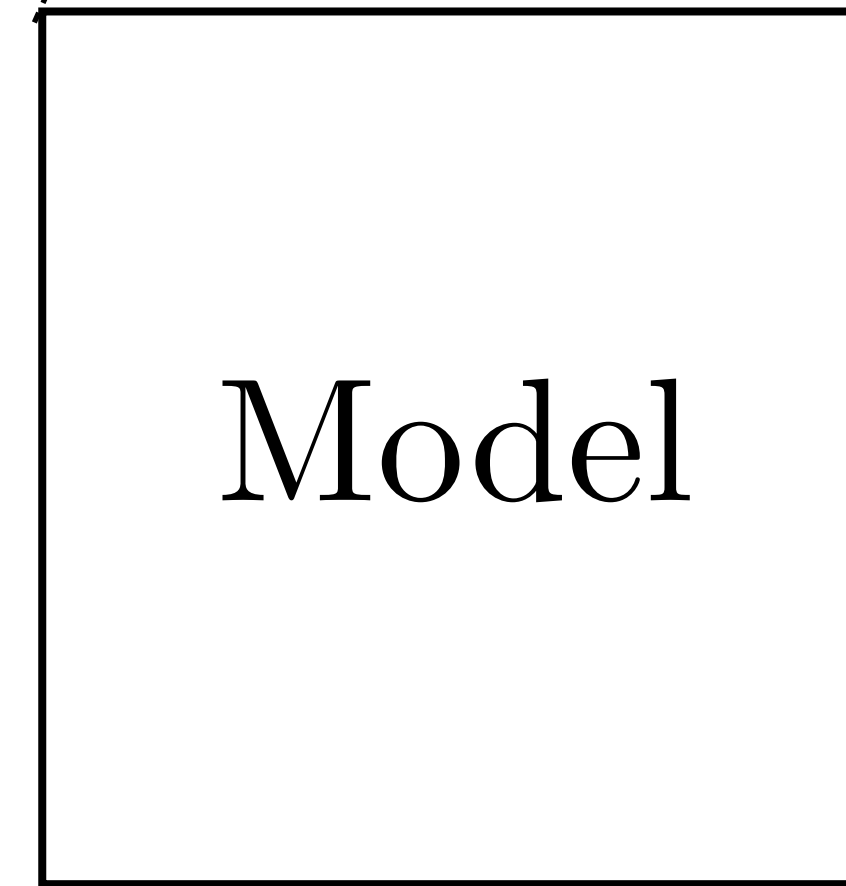
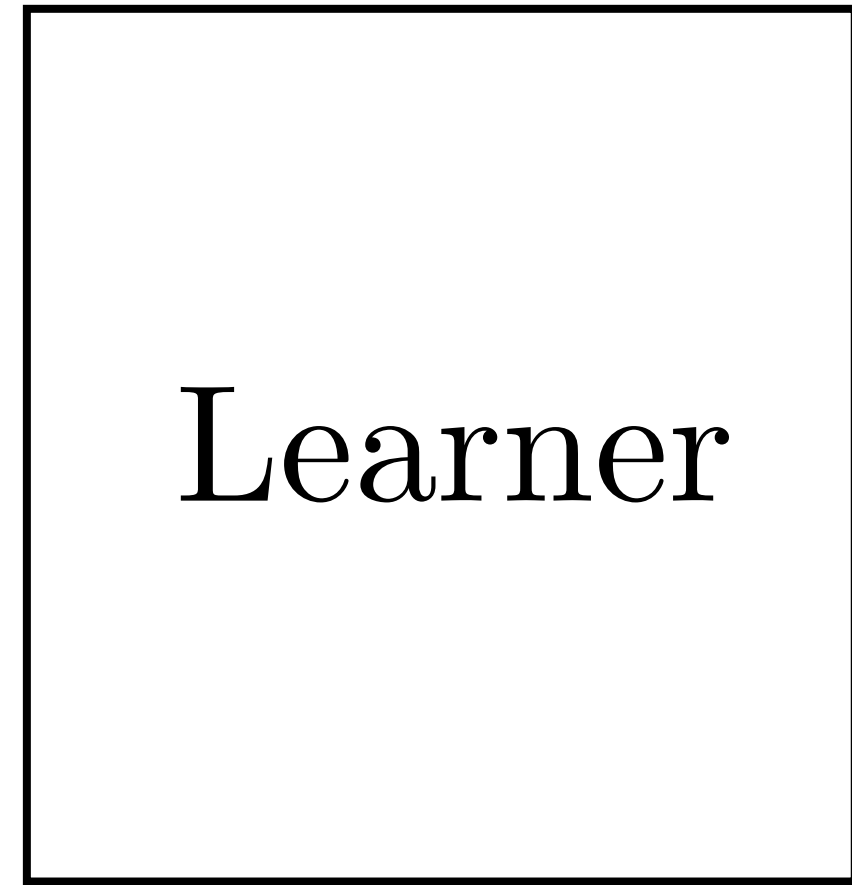
Data



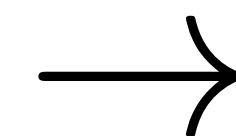
Learner



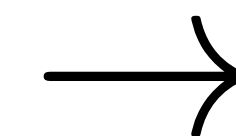
Model



Input



Model



Output

Inference

What does $\star$ do?

$$2 \star 3 = 36$$

$$7 \star 1 = 49$$

$$5 \star 2 = 100$$

$$2 \star 2 = 16$$

Learning / Training

What is $3 \star 5$?

Inference / Testing

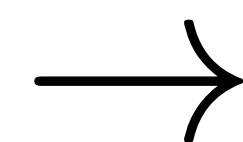
Training

$$2 \star 3 = 36$$

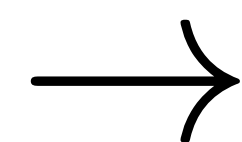
$$7 \star 1 = 49$$

$$5 \star 2 = 100$$

$$2 \star 2 = 16$$



Your
brain



$$x \star y \rightarrow (xy)^2$$

Testing

$$3 \star 5 \rightarrow$$

$$x \star y \rightarrow (xy)^2$$

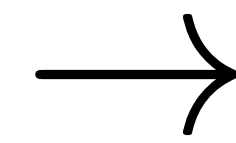
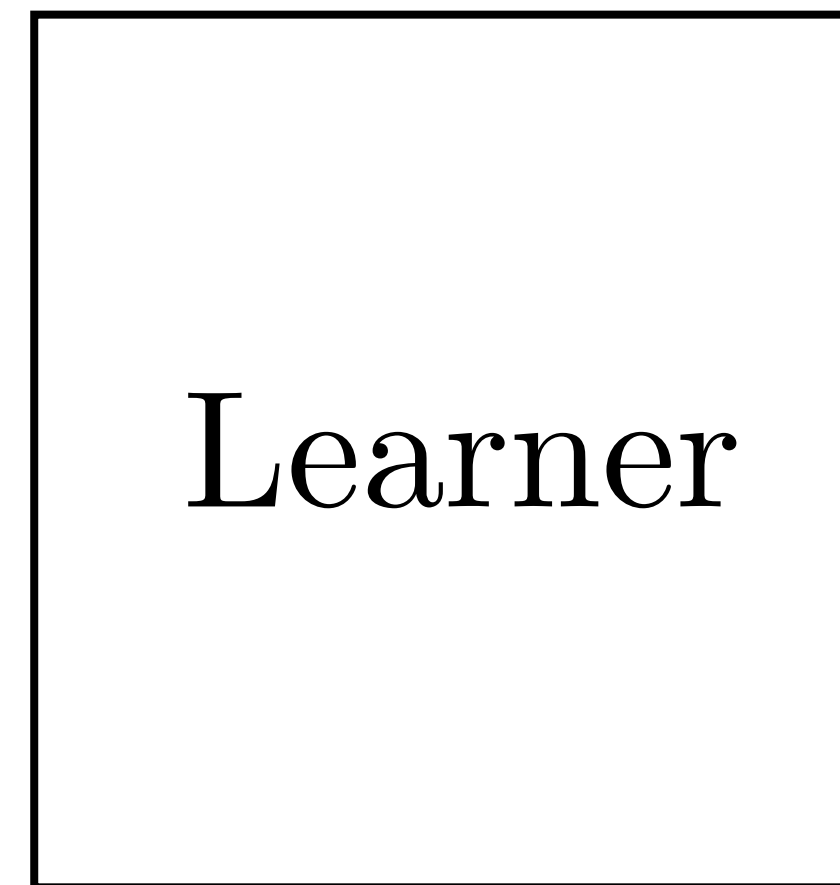
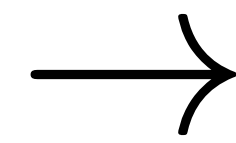
$$\rightarrow 225$$

Learning from examples

(aka **supervised learning**)

Training data

$\{\text{input}: [2, 3], \text{output}: 36\}$
 $\{\text{input}: [7, 1], \text{output}: 49\}$
 $\{\text{input}: [5, 2], \text{output}: 100\}$
 $\{\text{input}: [2, 2], \text{output}: 16\}$

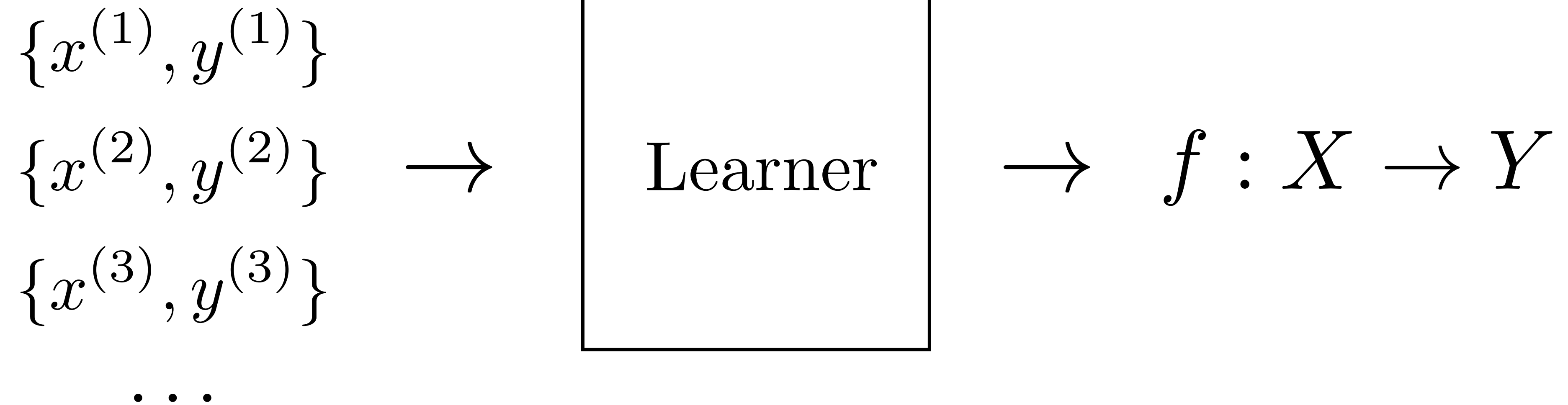


f

Learning from examples

(aka **supervised learning**)

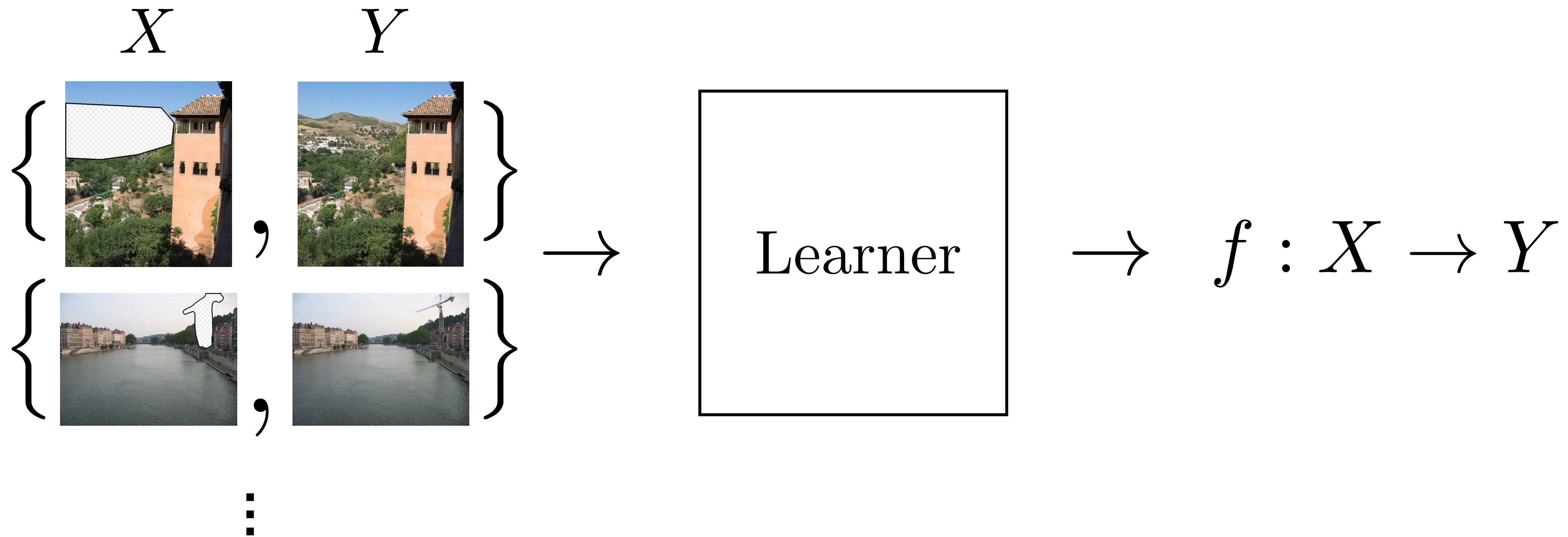
Training data



Learning from examples

(aka **supervised learning**)

Training data



Learning from examples

Case study #1: Linear least squares

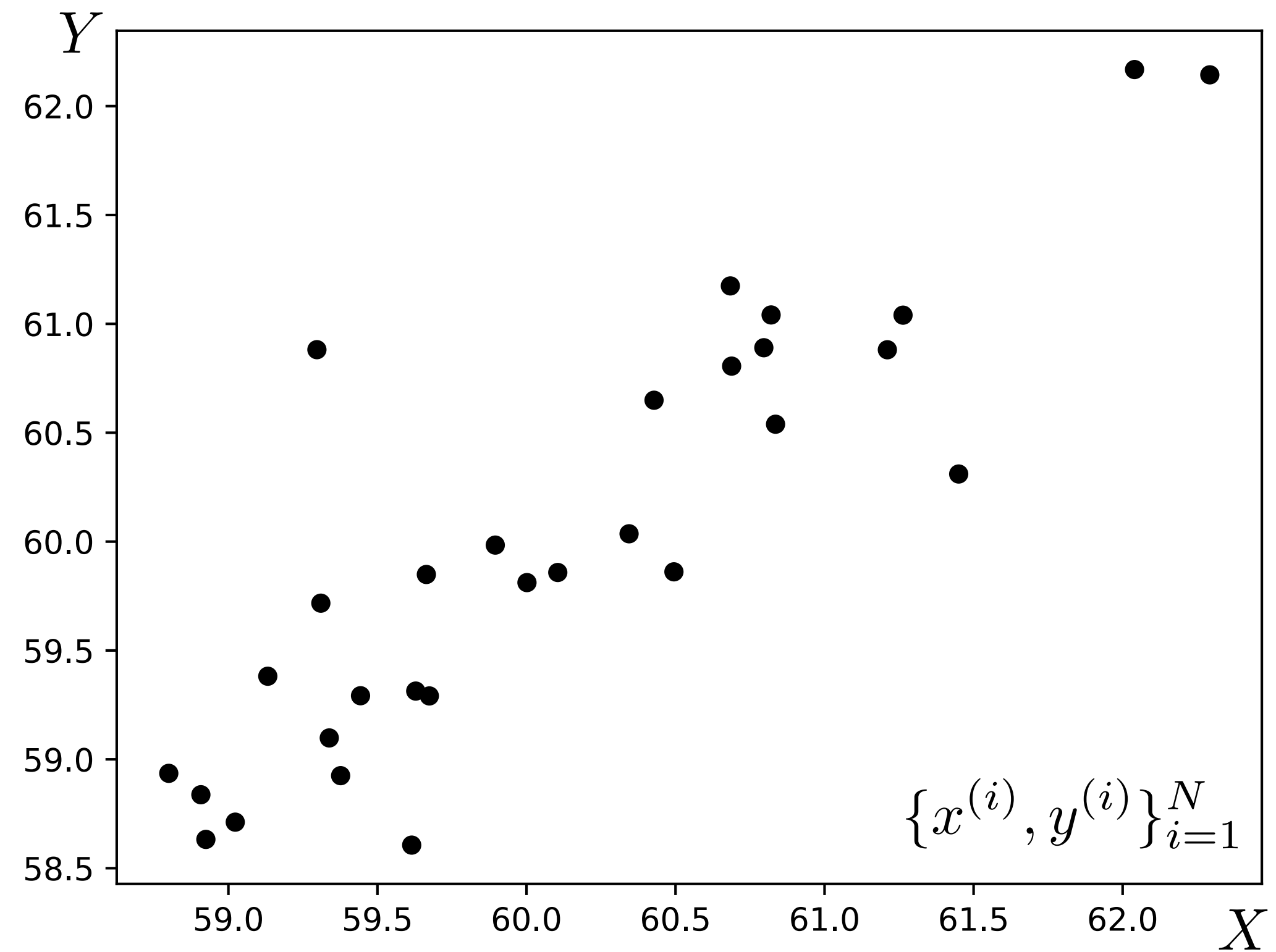
Hypothesis space

A vertical arrow pointing upwards from the text 'Hypothesis space' to the word 'Linear' in the title 'Case study #1: Linear least squares'.

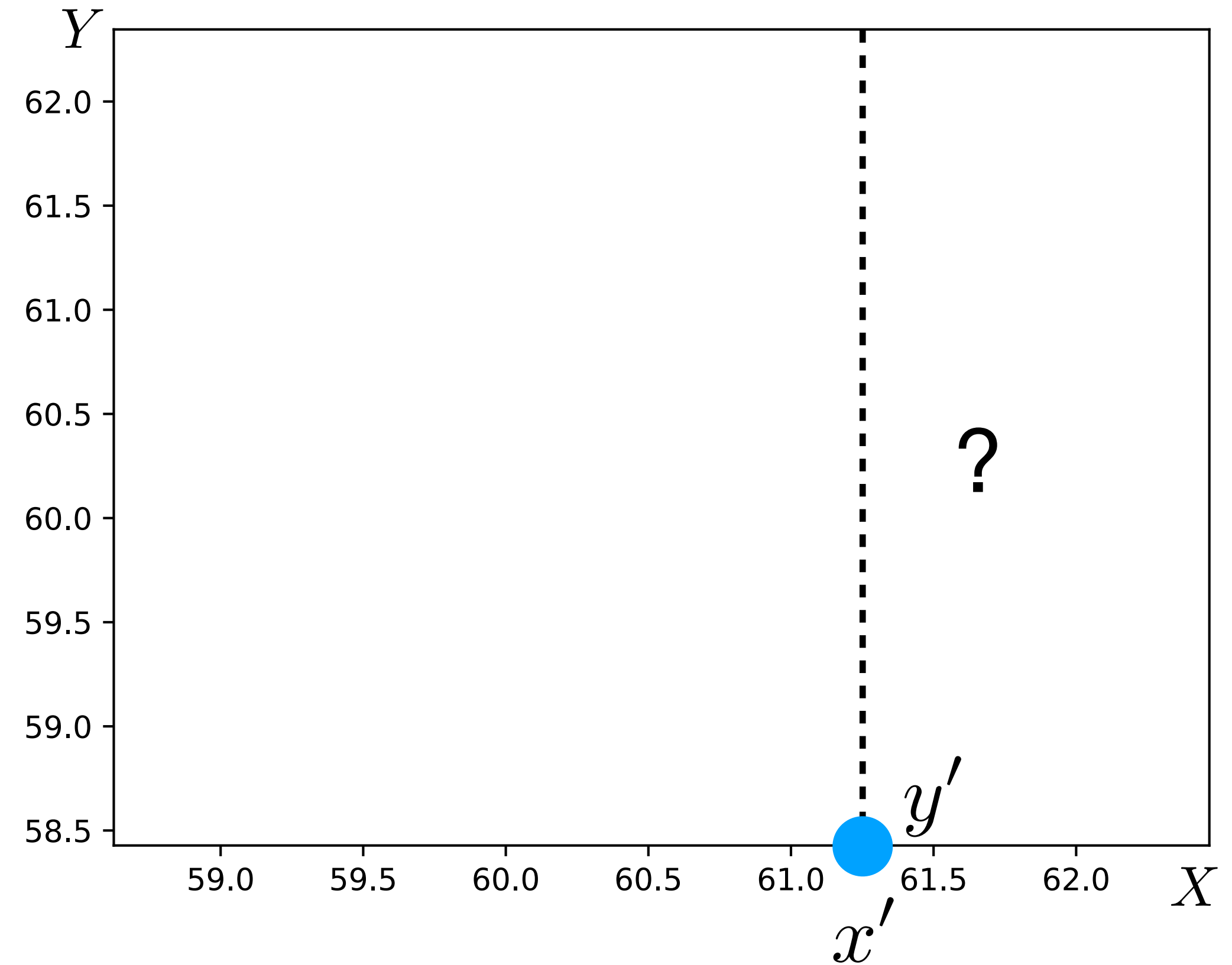
Loss term

A vertical arrow pointing upwards from the text 'Loss term' to the word 'least' in the title 'Case study #1: Linear least squares'.

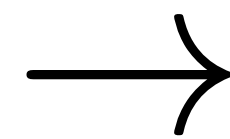
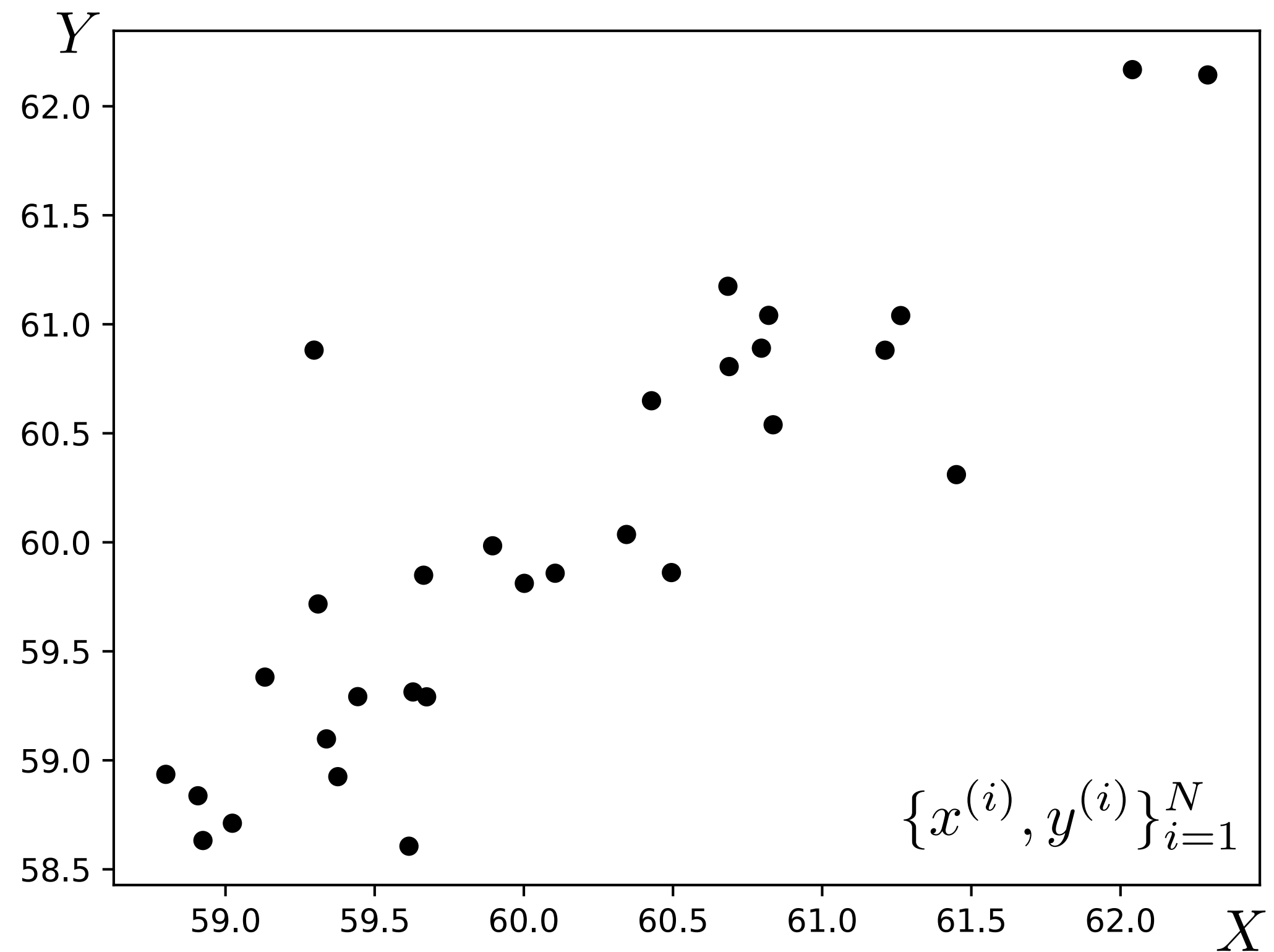
Training data



Test query



Training data



Learner

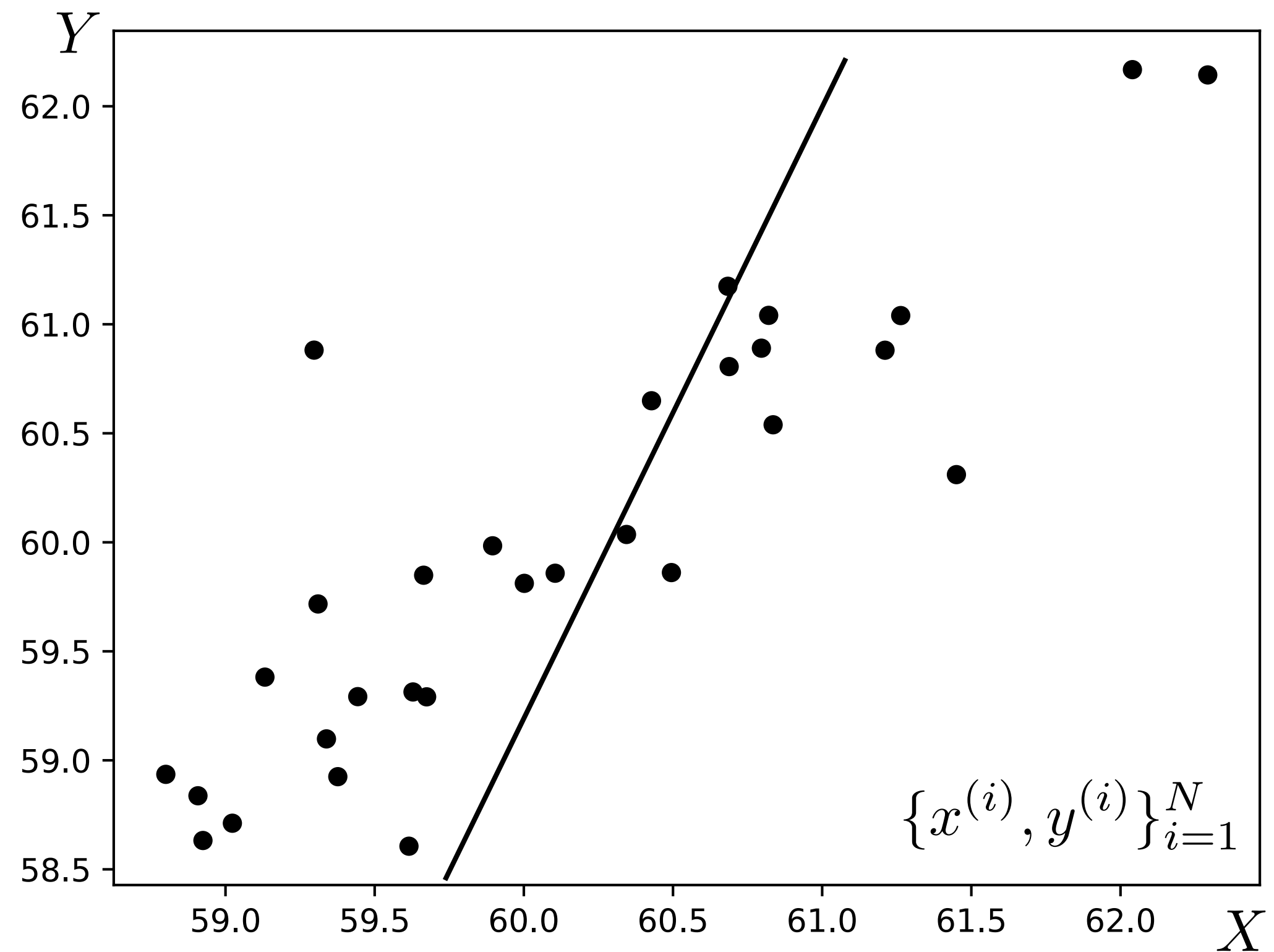


$$f_{\theta}(x) = \theta_1 x + \theta_0$$

Hypothesis space

The relationship between X and Y is roughly linear: $y \approx \theta_1 x + \theta_0$

Training data

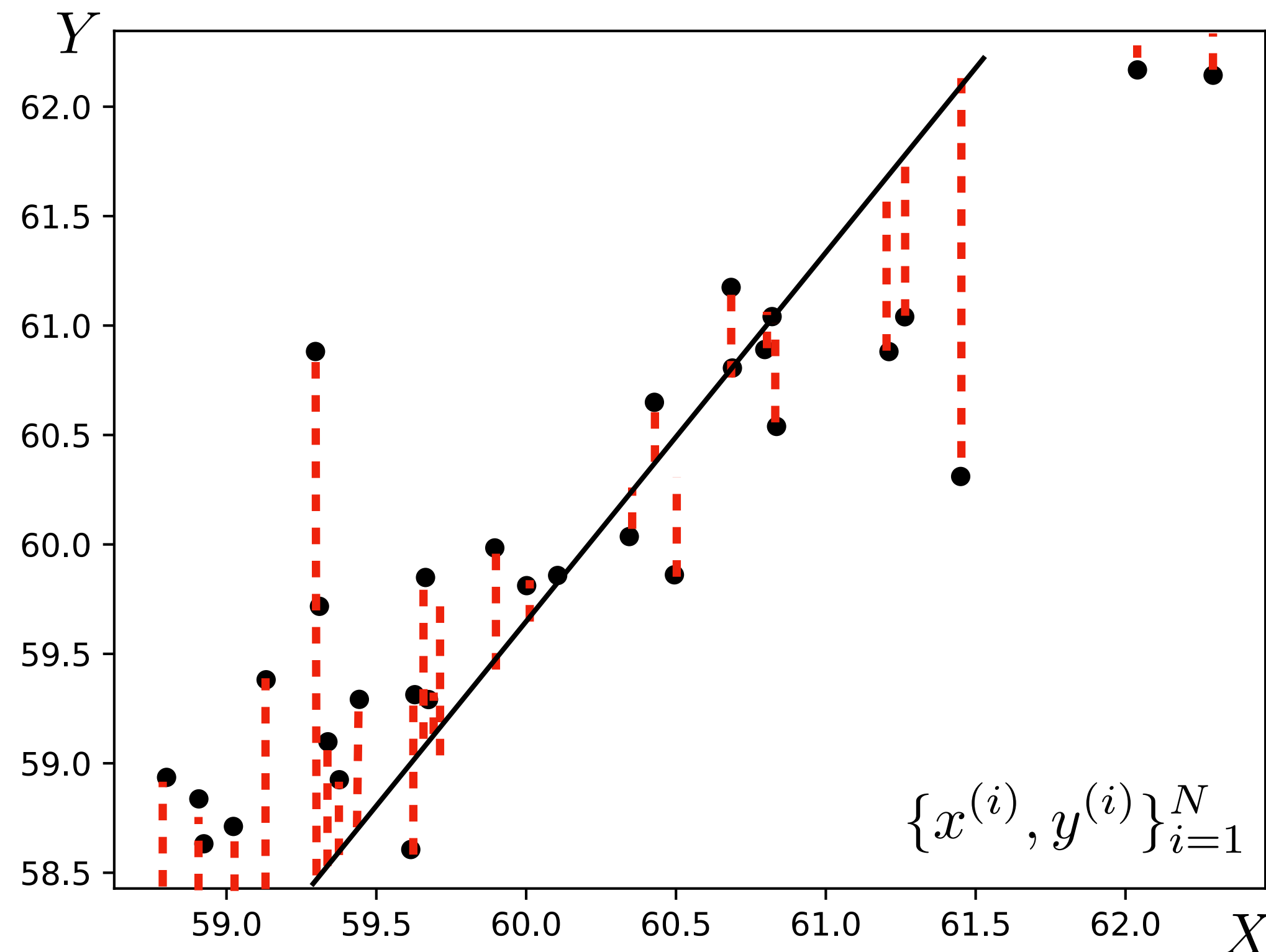


Search for the **parameters**, $\theta = \{\theta_0, \theta_1\}$, that best fit the data.

$$f_{\theta}(x) = \theta_1 x + \theta_0$$

Best fit in what sense?

Training data



Search for the **parameters**, $\theta = \{\theta_0, \theta_1\}$, that best fit the data.

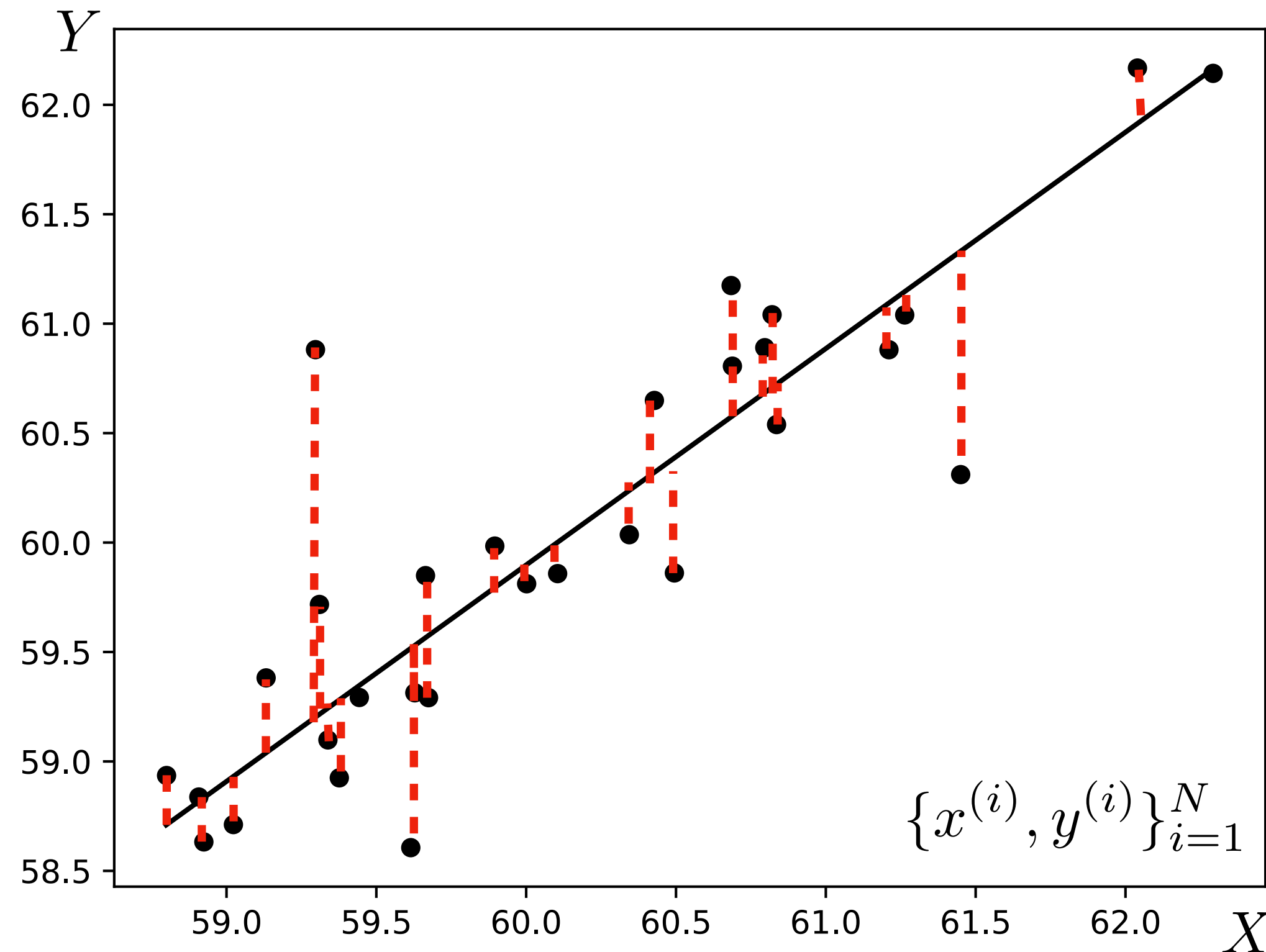
$$f_{\theta}(x) = \theta_1 x + \theta_0$$

Best fit in what sense?

The least-squares **objective** (aka **loss**) says the best fit is the function that minimizes the squared error between predictions and target values:

$$\mathcal{L}(\hat{y}, y) = (\hat{y} - y)^2 \quad \hat{y} \equiv f_{\theta}(x)$$

Training data



Search for the **parameters**, $\theta = \{\theta_0, \theta_1\}$, that best fit the data.

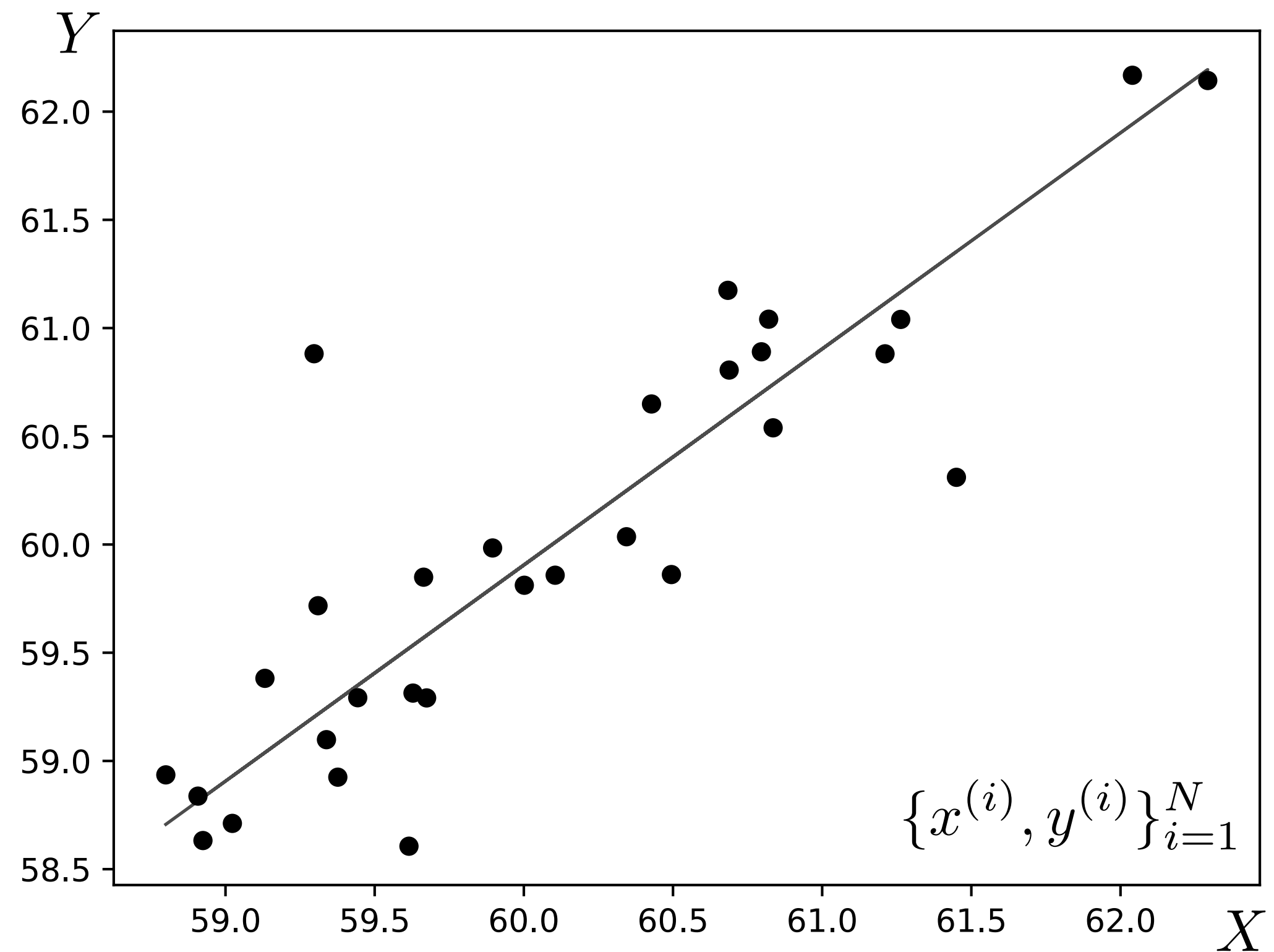
$$f_{\theta}(x) = \theta_1 x + \theta_0$$

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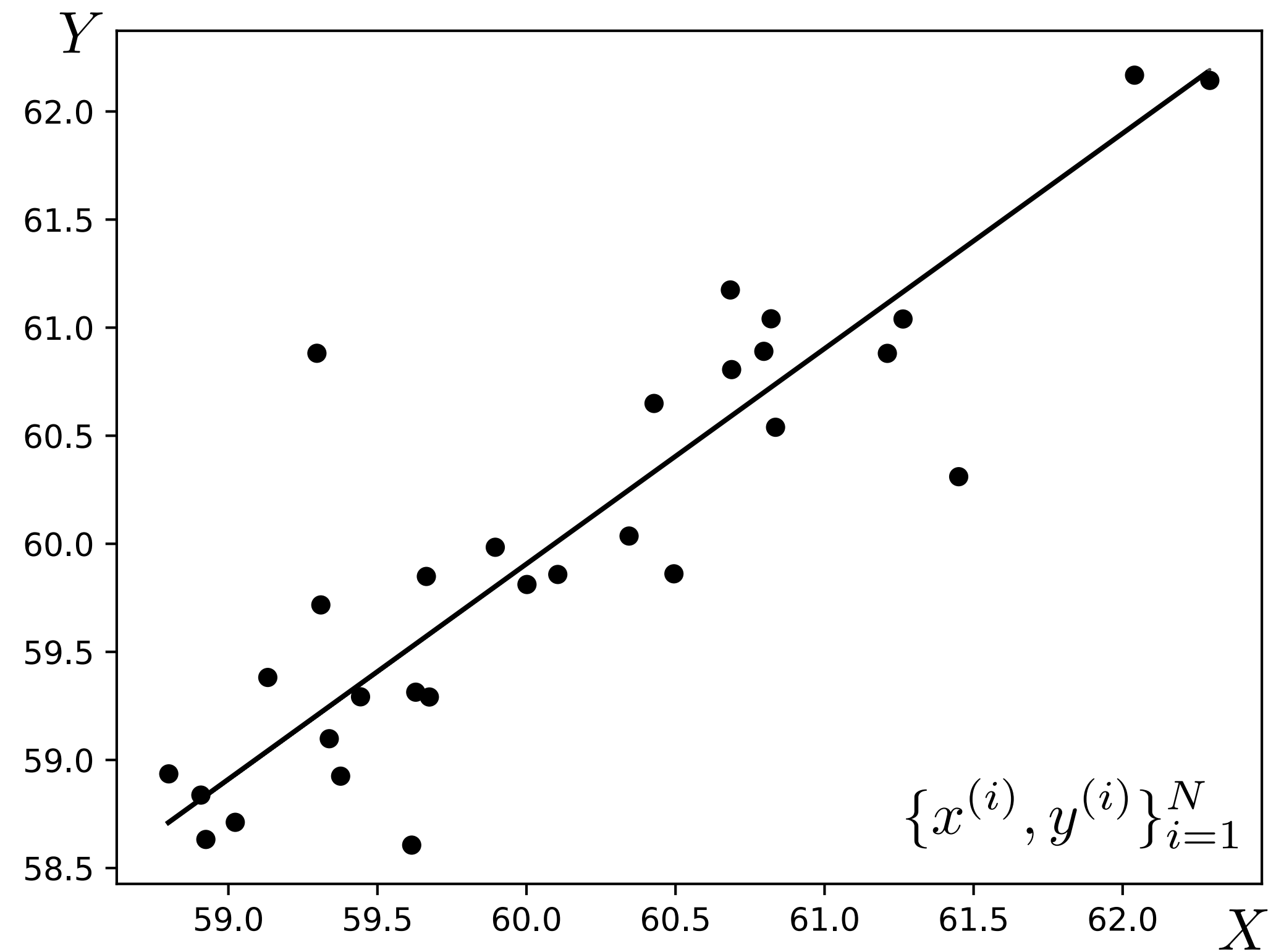
Training data



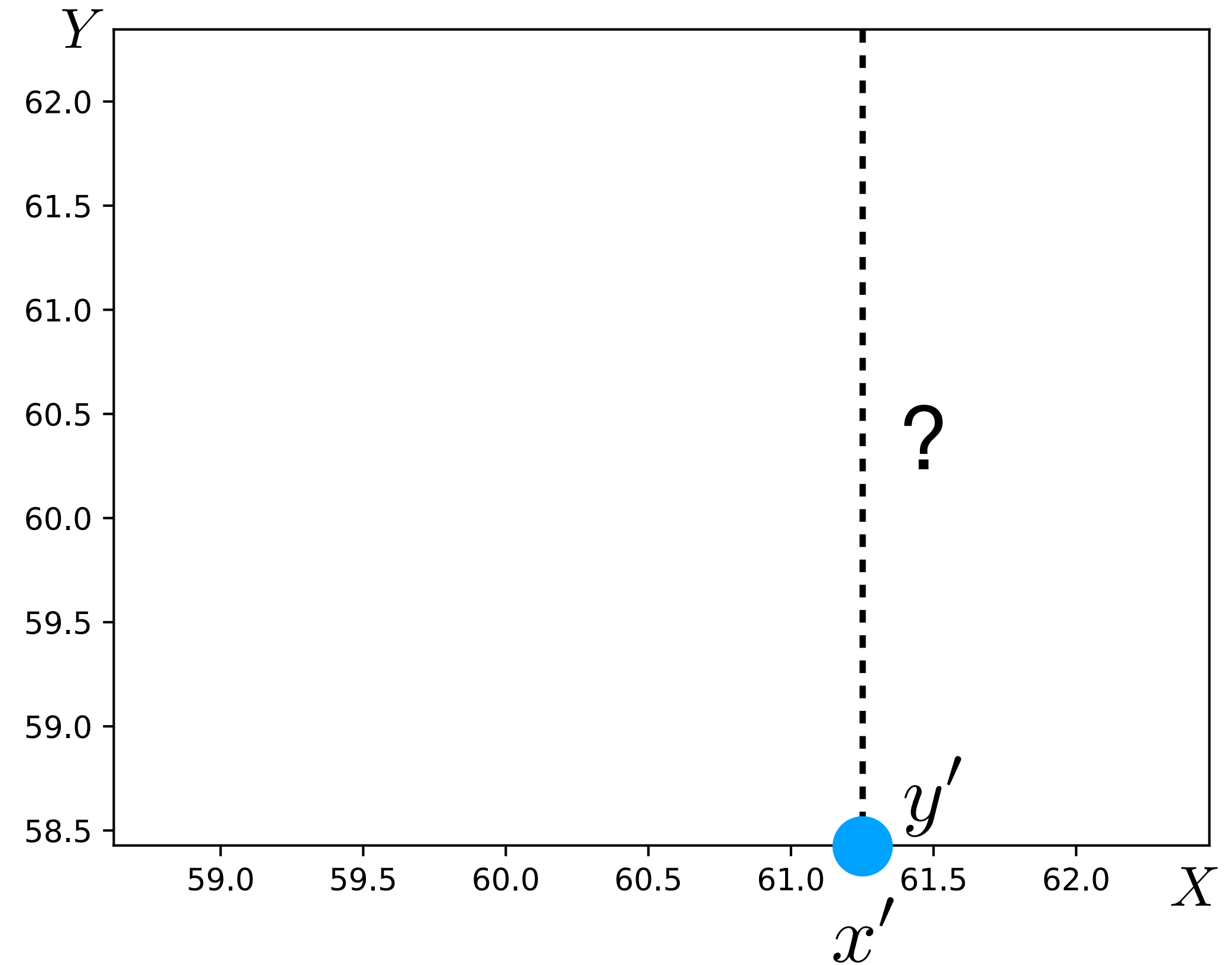
Complete learning problem:

$$\begin{aligned}\theta^* &= \arg \min_{\theta} \sum_{i=1}^N (f_{\theta}(x^{(i)}) - y^{(i)})^2 \\ &= \arg \min_{\theta} \sum_{i=1}^N (\theta_1 x^{(i)} + \theta_0 - y^{(i)})^2\end{aligned}$$

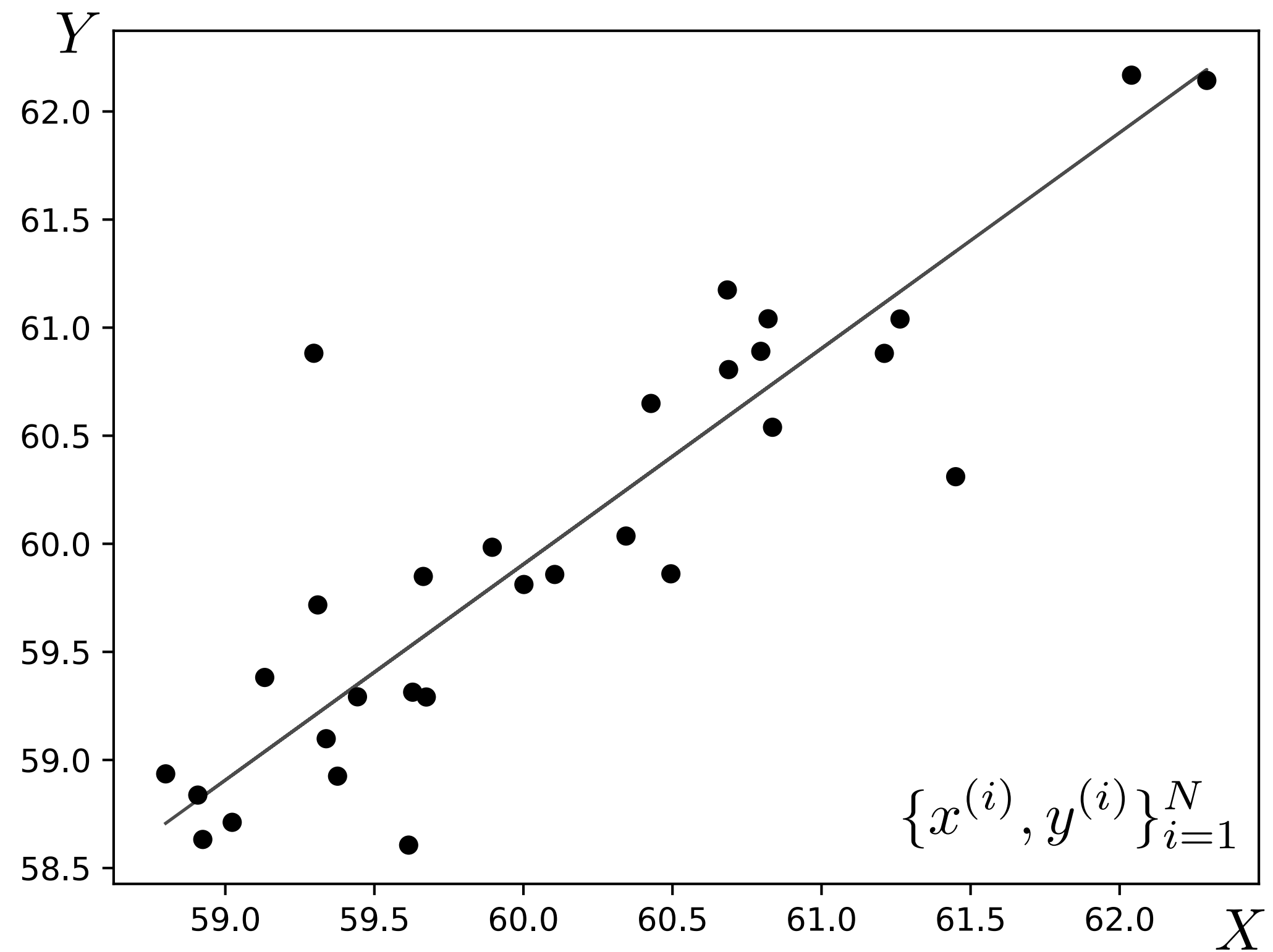
Training data



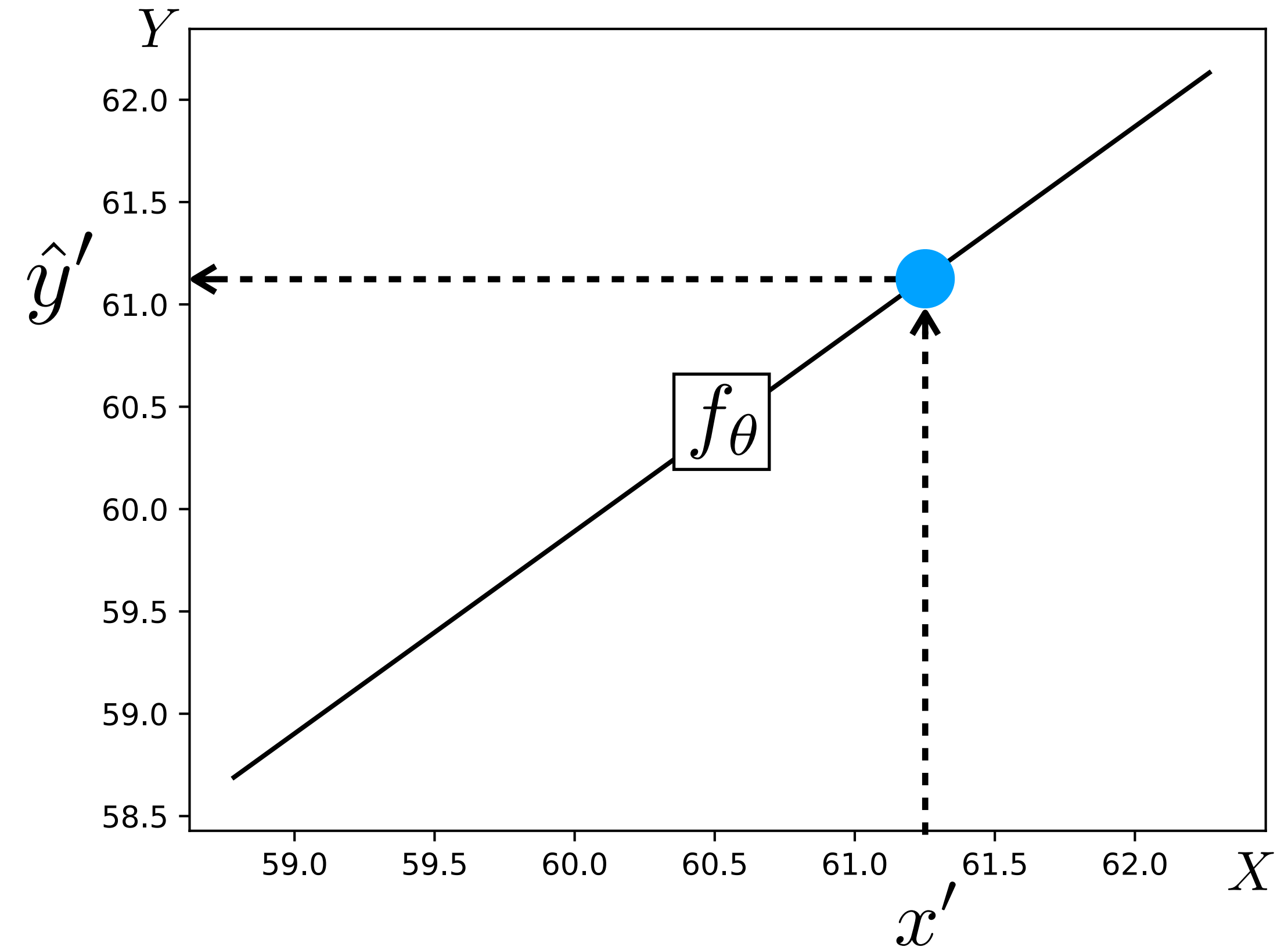
Test query



Training data



Test query

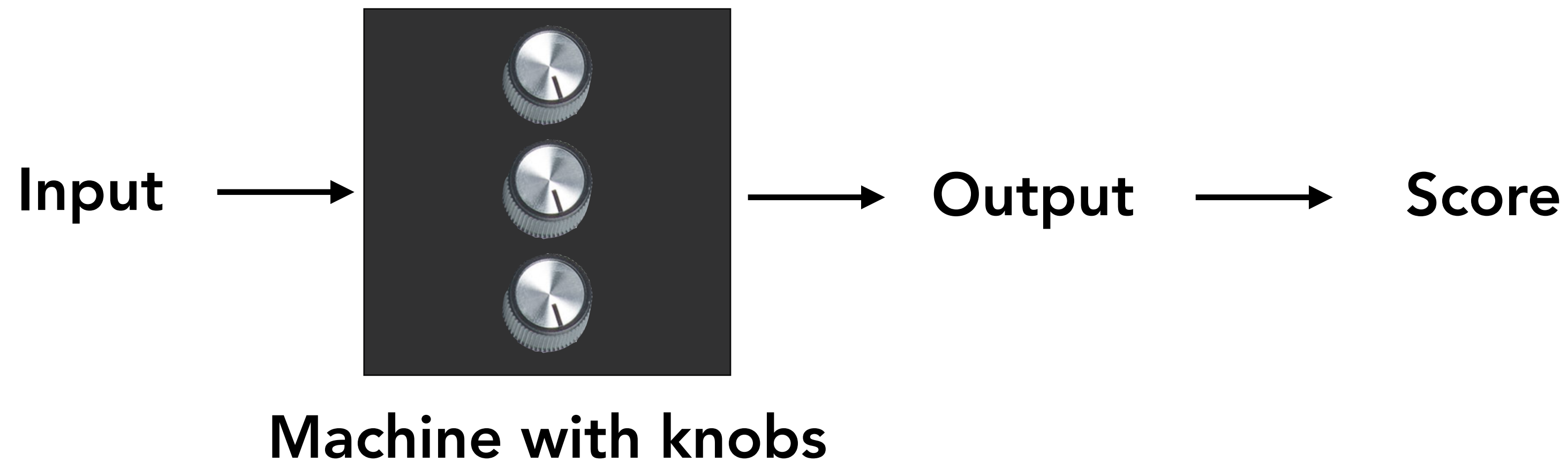


$$x' \dashrightarrow \boxed{f_\theta} \dashrightarrow \hat{y}'$$

How to minimize the objective w.r.t. θ ?

$$\theta^* = \arg \min_{\theta} \sum_{i=1}^N (f_{\theta}(x^{(i)}) - y^{(i)})^2$$

Use an **optimizer**!



Empirical Risk Minimization

(formalization of supervised learning)

Based on your data

Expected Loss

Linear least squares learning problem

$$\theta^* = \arg \min_{\theta} \sum_{i=1}^N (\theta_1 x^{(i)} + \theta_0 - y^{(i)})^2$$

Empirical Risk Minimization

(formalization of supervised learning)

Objective function
(loss)

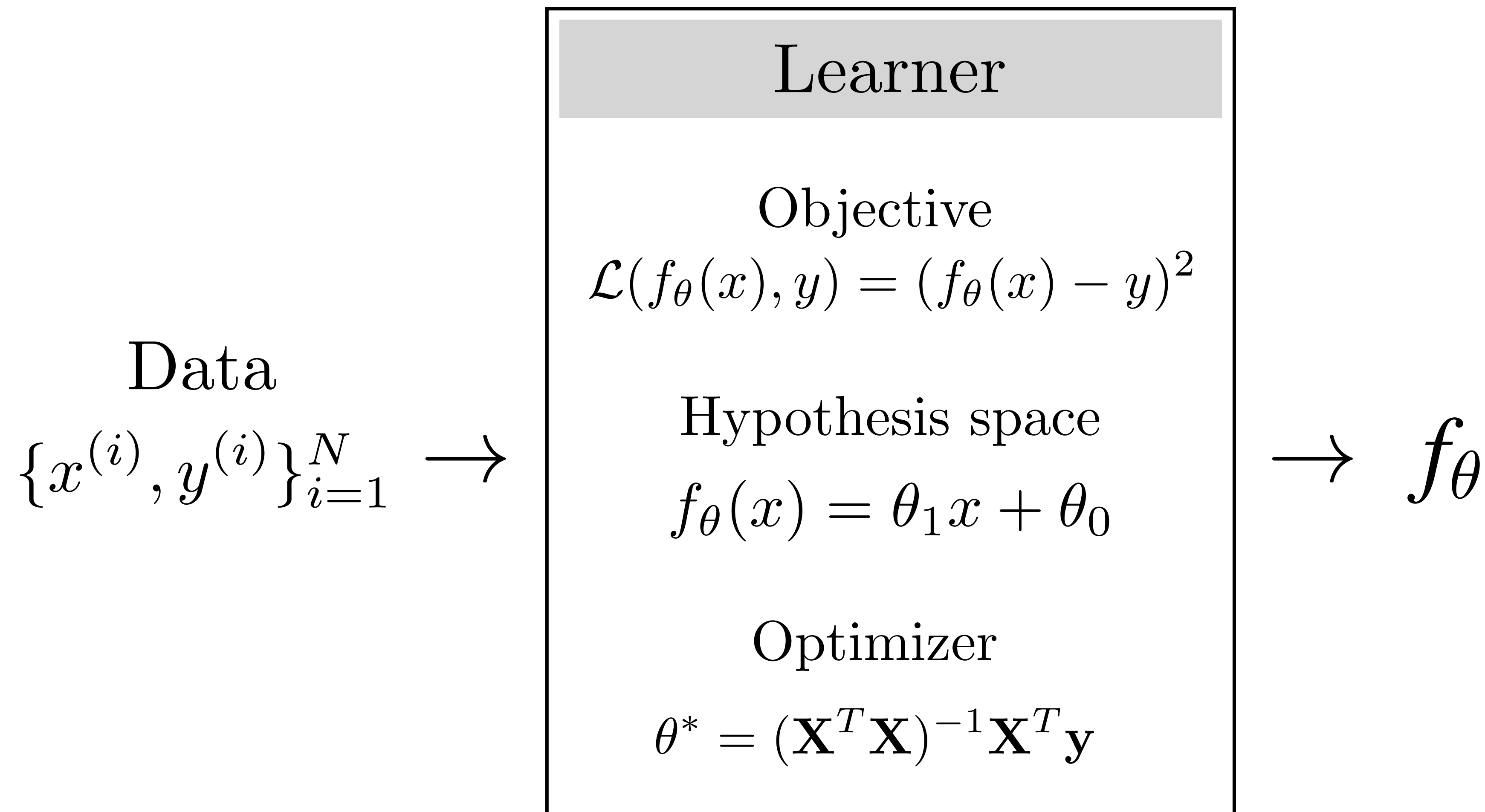
$$f^* = \arg \min_{f \in \mathcal{F}} \sum_{i=1}^N \mathcal{L}(f(\mathbf{x}^{(i)}), \mathbf{y}^{(i)})$$

Hypothesis space

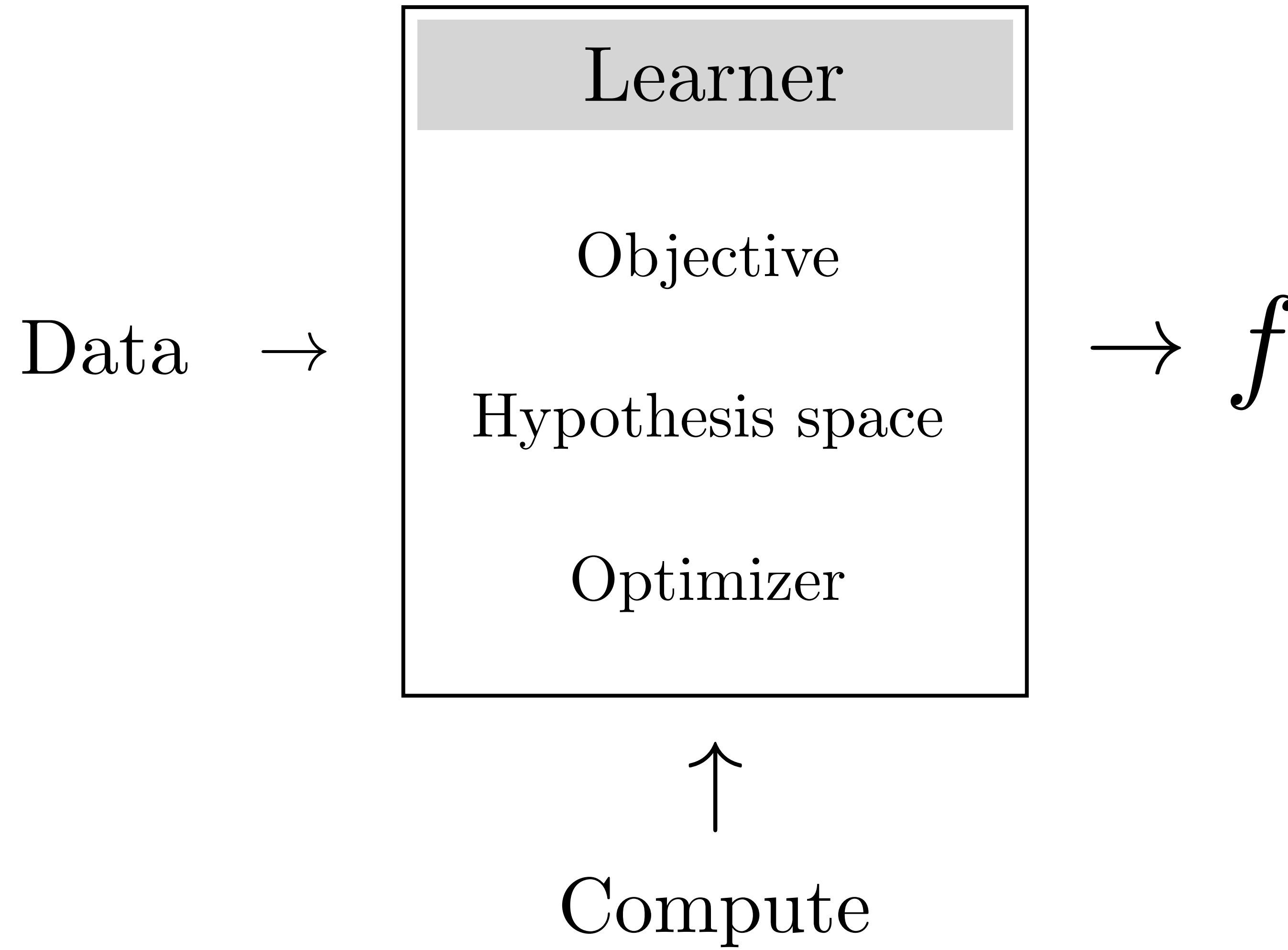
Training data

The diagram shows the equation for Empirical Risk Minimization. The term $f \in \mathcal{F}$ is annotated with 'Hypothesis space' and an arrow pointing to it. The summation $\sum_{i=1}^N$ is annotated with 'Objective function (loss)' and an arrow pointing to it. The function $\mathcal{L}(f(\mathbf{x}^{(i)}), \mathbf{y}^{(i)})$ is annotated with 'Training data' and two arrows pointing to $\mathbf{x}^{(i)}$ and $\mathbf{y}^{(i)}$ respectively.

Case study #1: Linear least squares

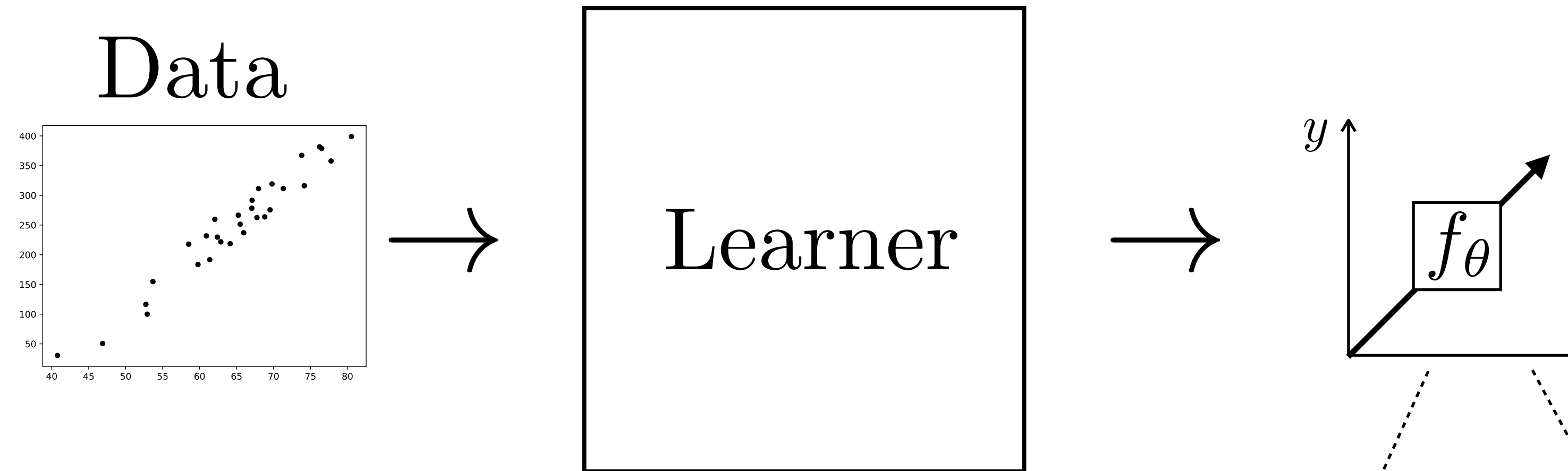


Key Ingredients

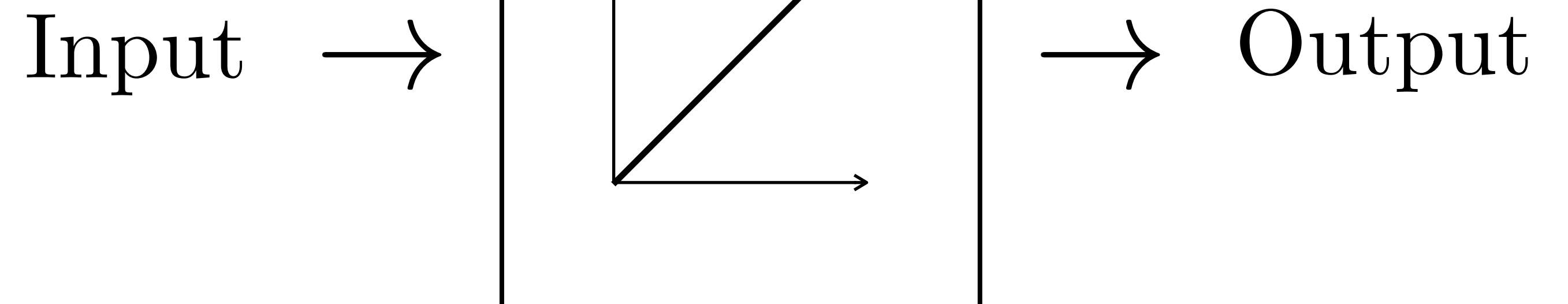


Hypothesis space 1: Linear functions

Training



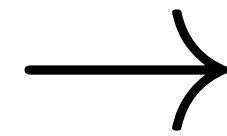
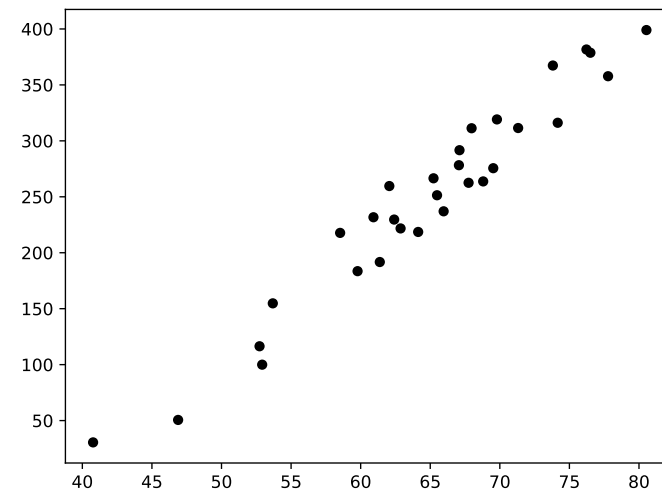
Testing



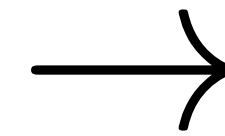
Hypothesis space 2: Python program induction

Training

Data



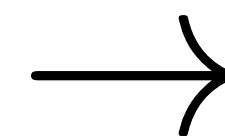
Learner



```
def predict(x):  
    y = 9.95*x - 395.43  
    return y
```

Testing

Input

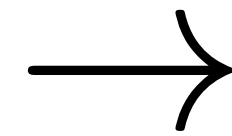
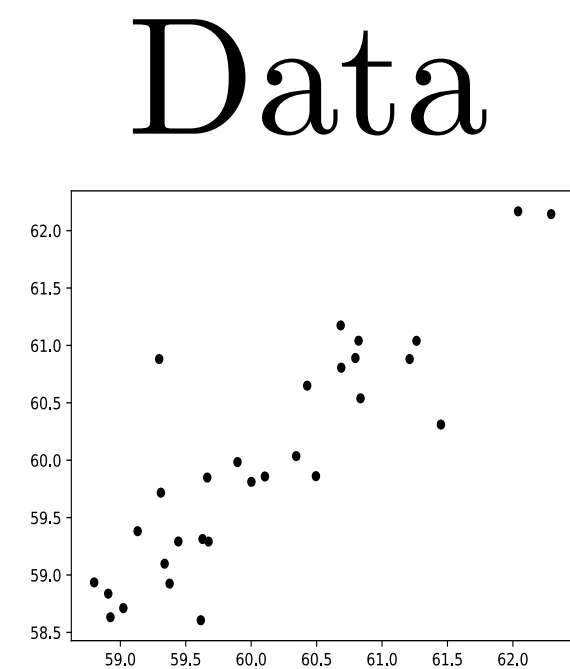


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def predict(x):  
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```

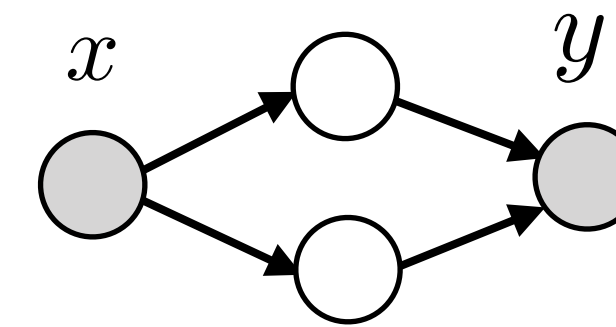
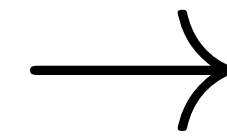
→ Output

Hypothesis space 3: Deep learning

Training

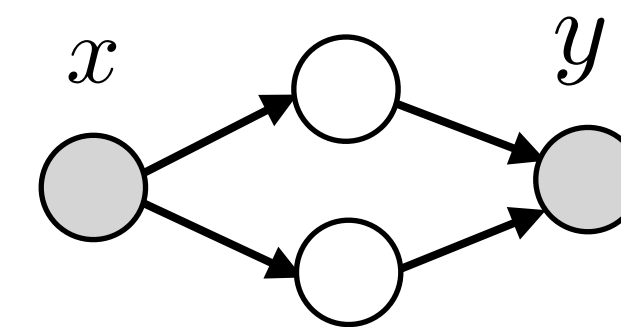
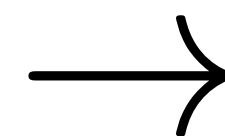


Learner



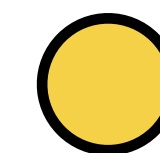
Testing

Input



Output

Space of all functions



Hypothesis space (haystack)



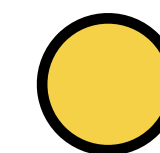
True solution (needle)

**Space we will
search**



Space of all functions

**Space we will
search**



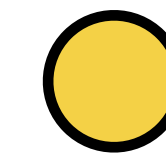
Hypothesis space (haystack)



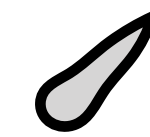
True solution (needle)

Deep nets

Space of all functions



Hypothesis space (haystack)



True solution (needle)

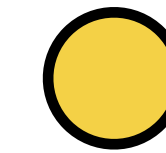
**Space we will
search**



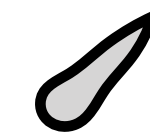
Linear functions

True solution is linear

Space of all functions



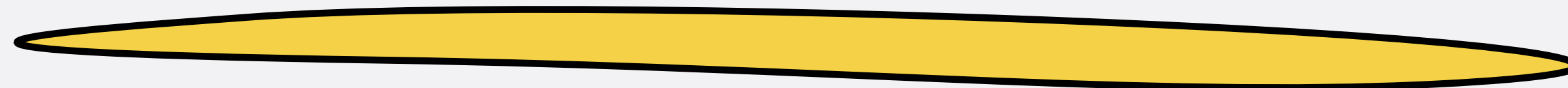
Hypothesis space (haystack)



True solution (needle)



**Space we will
search**



Linear functions

True solution is nonlinear